



MAKERERE UNIVERSITY
College of Computing and
Information Sciences
(CoCIS)

ANNUAL REPORT

2024
2025

Theme

“Fostering Academic Resilience and Long-Term
Achievement”



Message from the Principal



I am deeply appreciative of the additional extraordinary opportunity to serve as the Principal of the College of Computing and Information Sciences (CoCIS). I would like to provide a brief overview of our endeavours during the 2024/25 academic year.

This exceptional annual report is a key component of our reporting that highlights the work completed during the current academic year and serves as a powerful testament to the work that we are passionate about and perform to the best of our ability. Clearly, we have achieved enormous progress in the innovation and research domains, and we will continue to dominate these areas in the future with better and stronger concepts that will result in powerful,

innovative, and wonderful prototypes.

In January 2025, the College graduated 681 students: 12 doctorate candidates (7 females, 5 males), 73 master's candidates (26 females, 47 males), and 596 bachelor's graduates (251 females, 345 males).

The College honoured remarkable accomplishments, including three students who received the Africa 2024 Marketplace Simulation Award at the AFRICA2024 Conference in Egypt. The college also received a \$1.5 million grant from Google for the Ocular project, which uses artificial intelligence to improve malaria, TB, and cervical cancer detection. The AI Health Lab, led by Dr. Rose Nakasi, continues to develop breakthrough diagnostic tools, such as a phone-to-microscope adaptor, with support from Google and the National Institutes of Health. Additionally, the AirQo Project established the CLEAN-Air Africa Network, which focuses on air quality control with low-cost sensors.

Our College's Artificial Intelligence (AI) Health Lab also hosted its inaugural AI in African Health Conference on November 28, 2024, at the Imperial Royale Hotel in Kampala. The conference, titled "Building a Strong Ecosystem for AI in African Health Systems," sought to investigate the revolutionary potential

of AI in addressing Africa's healthcare concerns. This event brought together professionals from academia, healthcare, industry, and government to discuss cutting-edge AI-powered technologies for diagnostics, treatment optimization, and disease prediction. The conference stressed the importance of addressing varied healthcare requirements, promoting multidisciplinary collaboration, and developing long-term strategies for AI policy, legislation, and ethical application.

Our College recently welcomed Dr. Alo Oluwaseun Olubisi, a prominent lecturer from Nigeria, who will volunteer for two years in teaching, research, mentoring, and knowledge exchange. Dr. Olubisi, assigned to the Department of Computer Science, has settled in and is now contributing to the college's academic and research goals. I presented her to our PhD students to encourage collaboration on major research initiatives, as her involvement in these vital areas is expected to boost productivity and provide significant advantages to our College. Her nomination underlines our College's commitment to creating academic ties across Africa, strengthening capacity, and promoting creative research and knowledge exchange across the continent.

The Uganda Communications Commission (UCC), in collaboration with our College, successfully concluded a digital skills training program in 15 rural regions, with 2,643

participants. This effort, financed by the Uganda Communications Universal Service Access Fund (UCUSAF), sought to improve digital literacy in underserved and unserved communities between April 1 and June 19, 2024. The training covers key ICT skills such as computer and mobile phone use, Microsoft Office software, internet literacy, online collaboration, e-government services, and online safety. While the program was generally successful and received excellent comments, a number of issues arose. For example, many participants, notably in the Kotido and Bundibugyo districts, had challenges with device ownership. Despite UCC distributed over 150 smartphones, more than half of them were either broken or unusable, owing to a lack of spare parts and technical skills for repairs. The training was well-received since participants saw it as immediately related to their regular business activity. For example, in Amudat, participants could utilize their new skills to compare cattle prices in Kenya and Uganda, assisting with market judgments. The program drew more women than men in seven areas, with participation rates ranging from 51.5% to 69.5%. Participants indicated tremendous excitement for the training and asked more sessions tailored to their specific requirements.

During our freshers' orientation, I encourage them to maintain a positive attitude and avoid being persuaded by negative influences. "Stay away from negative energy, such as gossiping or undermining others," I

told me. "Positivity energizes your spirit and improves your knowledge, skills, and attitude." A pessimistic attitude will not get you far; but, positive thinking and kindness will. I also underlined that student should take control of their lives and avoid the traps of over-reliance on mobile phones for happiness. "True happiness is derived from within from your accomplishments, identity, and activities. I also reminded them how fortunate they were to be attending university; very few people have this chance, and emphasized the importance of striking a balance between academics and social life. "It is acceptable to attend parties, but not at the price of your education. "Remember, your primary responsibility here is academic success," I said. "Remind yourself every day that your ultimate goal is academic success. At the same time, enjoy your life by playing sports and visiting friends, but always prioritize your studies."

To Improve teaching and learning, the College has made substantial investments in pedagogical initiatives. We have heavily invested

in training and equipping our faculty over preceding six years, culminating in a robust and ready faculty capable of placing students at the centre of learning.

Finally, contemporary population fluctuations in Africa, particularly in Uganda, have yielded a demographic dividend. The majority of our population is young, aged 14 to 45, and Uganda has a fairly young population. This has both benefits and drawbacks. Some of the primary advantages include a technically knowledgeable and sophisticated young population, being in the economic productive age range, being active, innovative, and creative, being ready to work, being thrilled about market opportunities, and so on. Our College is therefore ready to deliver ICT leadership and training to this youthful generation.

Thank you very much for reading our Annual Report and when you are in Makerere University stop by and visit this amazing college.

We, as a college, are very committed to be the leading regional hub for Data Science, AI research and sensor technology.

Editor

Prof. Tonny J. Oyana
College Principal

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Table of CONTENTS

Message from the Principal.....	1
1.0 INTRODUCTION.....	6
1.1 The College of Computing and Information Sciences (CoCIS).....	6
1.2 The School of Computing and Informatics Technology (SCIT).....	6
1.3 The East African School of Library and Information Science (EASLIS)	6
1.4 Vision, Mission and Core values.....	7
1.5 College Leadership.....	8
2.0 TEACHING AND LEARNING.....	11
2.1 Programs offered at SCIT.....	11
2.2 Programs are offered at EASLIS.....	12
2.3 Students' Enrolment.....	12
2.4 Fresher's Orientation 2024/2025.....	14
2.5 Graduation output for the Mak 75th Graduation.....	15
2.6 Internship/Field Attachment.....	16
2.6.1 Field Attachment (FA) report from the EASLIS.....	16
2.6.2 Field Attachment (FA) report for the (SCIT).....	17
2.7 Curriculum Review and Accreditation status.....	18
3.0 RESEARCH AND INNOVATION.....	20
3.1 Undergraduate Students Win Africa 2024 Marketplace Simulation Award.....	21
3.2 Leveraging Artificial Intelligence for Disease Screening.....	21
3.2.1 Adapter Design and Fabrication.....	22
3.2.2 Malaria Diagnosis App.....	23
3.2.3 Data Collection App (Malaria and Cervical Cancer).....	23
3.2.4 Microscopy Teaching Aid.....	24
3.3 Air quality Monitoring Devices for Africa.....	24
3.3.1 AirQo second edition of the CLEAN-Air Forum in Lagos, Nigeria.....	26
3.3.2 Deputy Assistant Secretary for West Africa in the US. Department of States Bureau of African Affairs H.E Will Stevens Visits AirQo's Air Quality Monitoring Efforts in Nairobi, Kenya.....	26
3.4 Publications.....	27
3.5 Research Grants.....	36
4.0 COMMUNITY OUTREACH, KNOWLEDGE TRANSFER AND PARTNERSHIPS.....	40
4.1 Community Outreach.....	40

4.1.1 UCC and Mak- CoCIS Train 2,643 in Rural Areas to Boost Digital Literacy.....	40
4.1.2 Mak Ocular built capacity for laboratory technicians in health facilities	41
4.2 Workshops, seminars and conferences.....	44
4.2.1 Workshop on AI for students and staff participating in the Erasmus+ e program.....	44
4.2.2 Workshop on Integration of AI in Library and Information Sciences Education.....	44
4.2.3 Pedagogy Workshop.....	45
4.2.4 National celebration of D.E.A.R. Day.....	46
4.2.5 Makerere AI Health Lab and Ocular dissemination activities.....	46
4.3 Visitors.....	50
4.3.1 FPAIC Oversight Visit.....	50
4.3.2 Director of Africa Initiative visits CoCIS Labs.....	51
4.3.3 Germans seek collaboration.....	52
4.3.4 Kakiri Senior Secondary students visit Mak AI Health Lab.....	53
4.3.5 Nabinsunsa Girls Secondary School visit.....	53
5.0 THE STAFF ENVIRONMENT.....	54
5.1 Human Resources.....	54
5.1.2 Staff Promotions, Appointments and Confirmations in 2024.....	54
5.1.3 Volunteer Lecturer received from Nigerian Technical Aid Corps (TAC) program.....	56
5.1.4 Recognitions and Awards.....	56
5.1.4.1 Retired CoCIS Registrar and Accountants Honored.....	56
5.1.4.2 Ten staff get Vice Chancellor's Research Excellence Awards 2024	57
5.1.4.3 Dr. Rose Nakasi Awarded during Trend Setter Business forum.....	58
5.1.4.4 Staff establishment structure as of December 2024	60
5.1.4.5 Details of Academic Staff Qualifications and research interests.....	63
5.1.4.6 Administrative and Support staff.....	73
6.0 RECENT DEVELOPMENT.....	75
6.1 Launch of the Mak AI -Health Lab.....	75
6.2 DS-I Malaria Project Launched.....	76
6.2.1 Center for Innovations and Professional Skills Development (CiPSD)	77
7.0 FINANCIAL REPORT.....	81
Budget and Budget Performance-6 Months.....	81
8.0 CoCIS IN THE MEDIA.....	83

1.0 | INTRODUCTION

1.1 The College of Computing and Information Sciences (CoCIS)

Makerere University, College of Computing and Information Sciences (CoCIS) is the main ICT training, research and with service center in Makerere University and the region. The University Council established it on 1st February 2011 through the merge of the Faculty of Computing and Informatics Technology (CIT) and the East African School of Library and Information Sciences (EASLIS).

The College has two schools with six academic departments.

1.2 The School of Computing and Informatics Technology (SCIT)

School of computing and Informatics Technology (SCIT) is a center for ICT at Makerere University dealing with providing students with knowledge, skills and expertise in areas of computing and informatics Technology. The school is committed to fostering a conducive environment for teaching and learning. It offers a number of programs including Doctoral Programs, Masters Programs, Post Graduate Diplomas and Bachelor's Programs. Throughout this year, the school's grant, research and teaching portfolio has grown, with major successes in teaching outputs, grant funding, and research outputs. The school of Computing and Informatics Technology comprises of four departments:

- Department of Computer Science
- Department of Networks
- Department of Information Technology
- Department of Information Systems

1.3 The East African School of Library and Information Science (EASLIS)

The school was established in 1963. EASLIS pioneered the training of Librarians and Information Scientists in the East African region who have diligently served in different capacities across the social stratum. School

programs started from Certificate and Diploma levels that later were upgraded to Degrees and Post Graduate studies. Currently, the School offers Bachelors, Masters' Degrees (both in Library science and Records and Archives management) and Ph.D. in Information Science, producing a range of skilled professionals with competencies in Library Science, Records and Archive Management, Digital Libraries and Repository Managers, System analysts, Researchers, Information Literacy facilitators and trainer Publishers, among others. East African School of Library and Information Science (EASLIS) has two departments namely:

- Department of Records and Archives and the
- Department of Library Sciences.

Each department is engaged in teaching, supervising field attachment; conducting research, consultancy and outreach programs in the respective sub-disciplines.

1.4 Vision, Mission and Core values



Vision

To be a thought leader of Knowledge generation for societal transformation and development.



Mission

To provide first class teaching, research and services in Computing and Information Sciences responsive to national and international needs.



Core values

- Commitment, (Efficiency and Effectiveness)
- Transparency (Honesty and Integrity)
- Vibrancy (Enjoyment, Healthy environment and Sporting life)
- Respect (Friendly and Gender Sensitivity)
- Responsiveness

1.5 College Leadership

Table 1 **Principal, Deans and Heads of Departments**
Name, & Title & Unit, Tel. Contact & Email Address

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College of Computing and Information Sciences

**Mr. Julius Lebo****Principal Human Resource Officer**

College of Computing and Information Sciences

**Ms. Angida Francesca Mugyema****College Bursar**

College of Computing and Information Sciences

**Ms. Anyango Jane****Principal Communications Officer**

College of Computing and Information Sciences

2.0 | TEACHING & LEARNING

At CoCIS we are committed to creating a dynamic and effective educational environment in the field of Computing and Information sciences. This year we were actively engaged in Hands-on Learning, where we incorporated practical labs and projects that encourage experimental learning. We also organized workshops and mentorship programs. We also conducted regular assessment of teaching methods and course materials through student feedback, peer evaluations, and self-reflection to continuously improve the educational experience.

2.1 Programs offered at SCIT

SCIT offers a total of 15 programs at bachelors, Masters, PhD and Postgraduate levels. Following the merger of Information Technology and Information Systems programs in 2017, the school remained with three out of four undergraduate programs. The bachelor of information systems (BIS) and the bachelor of information technology (BIT) programs were merged to form bachelor of information systems and technology program (BIST). The Information Technology department has no undergraduate program. The list of programs offered include:

1. Bachelor of Science in Computer Science
2. Master of Science in Computer Science
3. Postgraduate Diploma in Computer Science
4. PhD in Computer Science.
5. Bachelor of Information Systems and Technology,
6. Master of Science in Information Systems,
7. Postgraduate Diploma in Information Systems,
8. PhD in Information Systems
9. Masters in Information Technology (MIT)
10. PhD in Information Technology (Ph.D. in IT)
11. Post Graduate Diploma in Information Technology (PGDIT).
Post Graduate Diploma in Information Technology (PGDIT)
12. Bachelors in Software Engineering
13. Masters in Data communication and Software Engineering
14. Post graduate diploma in Data communication and Software Engineering
15. PhD in Software Engineering.

2.2 Programs offered at EASLIS:

EASLIS serves as a regional training institution for Library and Information Science equipping students with skills to work in any information institutions like Libraries, Archival Institutions, Registries, Records Offices, and Publishing houses, Book distribution Centers, Documentation Centers, Information Centers, Resource Centers etc.

Currently, EASLIS offers five programs:

1. Bachelor of Library and Information Science (BLIS)
2. Bachelor of Records and Archives Management (BRAM)
3. Master of Science in Information Science (MSC.INF.SCI)
4. Master of Science in Records and Archives Management (MSC.RAM)
5. Doctor of Philosophy in Information Science (PhD INF. SCI)

2.3 Students' Enrolment

CoCIS has a total of 2152 students enrolled for the academic Year 2024/2025.

Table 2 Enrollment Statistics Academic Year 2024/2025
Semester One by school, program and gender

A School of Computing and Informatics Technology

ENROLLED STUDENTS					
			Female	Male	Total
1	Postgraduate Diploma in Computer Science	Year 1	0	2	2
2	Master of Science in Computer Science	Year 1	10	34	44
		Year 1	4	8	12
3	Postgraduate Diploma in Information systems	Year 1	0	1	1
4	Master of Science in Information systems	Year 1	8	29	37
		Year 2	13	18	31
5	Postgraduate Diploma in Information Technology	Year 1	0	1	1

6	Master of Information Technology	Year 1	7	27	34
		Year 2	4	18	22
7	Postgraduate Diploma in Data communication and Software Engineering	Year 1	1	1	2
8	Master of Science in Data communication and Software Engineering	Year 1	4	30	34
		Year 2	4	17	21
9	Bachelor of Science in Computer Science	Year 1	57	91	148
		Year 2	56	93	149
		Year 3	17	61	78
10	Bachelor of Science in Software Engineering	Year 1	54	80	134
		Year 2	57	105	162
		Year 3	8	33	41
		Year 4	53	101	154
11	Bachelor of Information Systems and Technology	Year 1	1	7	8
		Year 2	100	180	280
		Year 3	33	84	117
12	Bachelor of Information systems	Year 3	0	3	3
13	Bachelor of Information Technology	Year 3	3	6	9
14	Doctor of Philosophy (SCIT)	Year 1	3	0	3
		Year 2	0	0	0
		Year 3	0	2	2
		Year 4	0	1	1
15	PhD in Computer Science	Year 2	0	2	2
16	PhD in Information Systems	Year 1	2	4	6
17	PhD in Information Technology	Year 3	0	1	1
Grand Total for SCIT (A)					1539

B East African School of Library and Information Science

ENROLLED STUDENTS					
			Female	Male	Total
18	Bachelor of Library and Information Science	Year 1	171	65	236
		Year 2	153	59	212
		Year 3	39	28	67
19	Bachelor of Records and Archives Management	Year 3	5	3	8
20	Master of Science in Information Science	Year 1	20	12	23
		Year 2	19	10	29
21	Master of Science in Records and Archives Management	Year 1	5	6	11
		Year 2	7	17	8
22	PhD in Information Sciences	Year 1	3	2	5
		Year 2	1	1	2
		Year 3	3	0	3
Grand Total for EASLIS (B)					613
OVERALL (A + B)					2152

2.4 Fresher's Orientation 2024/2025

The College welcomed freshers for the 2024/2025 academic year with an orientation session aimed at easing their transition into university life. Led by College Principal Professor Tony Oyana, the event emphasized positivity, responsibility, and the effective use of university resources. Heads of academic departments provided insights into various programs, while representatives from non-academic units, including Guidance and Counselling, the University Hospital, and Security, introduced students to essential support services. The College Registrar's Office guided students through registration processes, covering tuition payment and academic policies. The event concluded with a Question-and-Answer session, allowing freshers to seek clarification and better understand their new academic environment.



East African School of Library and Information Science



Dr. Peter Nabende hands over gifts to winners of the orientation raffle

2.5 Graduation output for the Mak 75th Graduation

At Makerere University's 75th Graduation Ceremony on January 13, 2025, the College of Computing and Information Sciences (CoCIS) presented a total of 681 graduands. These included 12 doctoral candidates (7 females, 5 males), 73 master's candidates (26 females, 47 males), and 596 bachelor's graduates (251 females, 345 males).



A section of undergraduate candidates at the Mak 75th Graduation



A section of Masters graduands



Some of the PhD graduands from CoCIS

2.6 Internship/Field Attachment

Students were actively engaged in Field activities supervised by academic staff. This practical experience is crucial for their professional development and complements their academic learning.

2.6.1 Field Attachment (FA) report from the EASLIS

The 2023/2024 Field Attachment preparations at the East African School of Library and Information Sciences involved issuing placement letters and securing practicum sites, though some students prioritized convenience over professional growth. Orientation sessions for students and staff were conducted before disbursement of funds, followed by a mid-term review on June 27,

2024, to assess progress. A post-field attachment workshop on October 5, 2024, evaluated supervision, assessed reports, and finalized results. Of the 43 participating students, two missed supervision due to communication issues, while two failed to submit reports despite reminders. Challenges included poor report-writing skills, budget constraints as more staff participated in monitoring than planned, and student tardiness. Issues with academic and host supervisors also led to report submission delays. There is need for improving communication, enforcing placement guidelines, and enhancing student accountability for future field attachments.



EASLIS members of staff at a post evaluation

2.6.2 Field Attachment (FA) report for the (SCIT)

The Field Attachment (FA) activities at the School of Computing and Informatics Technology for the 2023/2024 academic year followed a structured internship process undertaken by students. The activities were divided into three phases: Pre-placement, Field Supervision, and Post-Evaluation. The pre-placement phase involved planning, budgeting, student sensitization, and securing placements, with a total of 330 students and a budget of UGX 65 million. The school identified and partnered with 68 organizations through self-placement and industry engagement. Field supervision was conducted between May and July 2024, ensuring that students' performance was monitored and assessed.

The FA was successfully completed but with key challenges, such as students delaying internship start dates and delayed disbursement of supervision funds. Despite these hurdles, the evaluation of the FA program was done in October 2024. There was a slight shortfall in funds for field supervision, though most

activities were successfully implemented within budget. There is need for improving student participation and financial planning for future FA activities.

2.7 Curriculum Review and Accreditation status

Curriculum reviews are ongoing. Most programs were accredited with continuous improvements being made at different levels of approval as shown in the table 3 below.

Table 3 Status of Accreditation of Programmes of Study Tenable at the College of Computing and Information Sciences (CoCIS)

SN	Name of Programme of Study	Department	School	Previous Nche Accreditation Date	Current Stage In The Review/ Re-Accreditation Process
1	Bachelor of Library and Information Science	Library and Information Science	EASLIS	August 2022	Valid till August 2027
2	Bachelor of Science in Computer Science.	Computer Science	SCIT	March 2022	Valid till March 2027
3	Bachelor of Science in Software Engineering	Networks	SCIT	March 2022	Valid till March 2027
4	Bachelor of Information Systems and Technology.	Information Systems	SCIT	August 2017	Pending approval by the NCHE
5	Postgraduate Diploma in Computer Science	Computer Science	SCIT	August 2017	Valid till August 2026
6	Postgraduate Diploma in Data Communication and Software Engineering	Networks	SCIT	August 2017	Valid till August 2026
7	Postgraduate Diploma in Information Systems	Information Systems	SCIT	August 2019	Submitted for review to Senate

8	Postgraduate Diploma in Information Technology	Information Technology	SCIT	August 2020	Submitted for review to Senate
9	Master of Science in Computer Science	Computer Science	SCIT	August 2021	Valid till August 2026
10	Master of Science Data Communication and Software Engineering	Networks	SCIT	August 2021	Submitted for review to Senate
11	Master of Information Technology	Information Technology	SCIT	August 2021	Submitted for review to Senate
12	Master of e-Governance Systems and Technology	Information Technology	SCIT	New Programme - 2023	Approved by NCHE in 2024
13	Master of Science in Information Science	Library and Information Science	EASLIS	August 2022	Valid August 2030
14	Master of Science in Records and Archives Management	Records and Archives Management	EASLIS	August 2019	Approved in February 2025
15	Master of Science in Information Systems	Information Systems	SCIT	August 2019	Submitted for review to Senate
16	Doctor of Philosophy in Computer Science	Computer Science	SCIT	March 2010	Pending approval by the NCHE in 2025
17	Doctor of Philosophy in Software Engineering	Networks	SCIT	March 2009	Pending approval by the NCHE Council 2025
18	Doctor of Philosophy in Information Systems	Information Systems	SCIT	August 2019	Valid till August 2029
19	Doctor of Philosophy in Information Technology	Information Technology	SCIT	August 2010	Pending approval by the NCHE in Feb 2025
20	Doctor of Philosophy in Information Science	Library and Information Science	EASLIS	August 2016	Approval by the NCHE Council in February 2025
21	Doctor of Philosophy in Records and Archival Science	Records and Archives Management	EASLIS	New Programme - 2023	Approval by the NCHE in Feb. 2025

3.0 | RESEARCH AND INNOVATION

The College has made significant strides in research and innovation, establishing itself as a key player in the field. It consistently produces high-quality research, evidenced by numerous publications in reputable journals and conferences. Faculty and students engage in cutting-edge areas such as artificial intelligence, data science, cybersecurity, human-computer interaction, and software engineering. Collaborative, interdisciplinary projects have led to innovative solutions addressing real-world challenges, further strengthening the college's reputation for excellence.

E-learning

This year, 90.3 % of the courses at SCIT were blended with e-Learning. Students especially first years were trained in e learning on 14th of August 2024 These students were able to enroll I for courses at SCIT on line and access coursework materials and online course assessments.

Seventy one percent of staff at SCIT were technically supported by the e-learning coordinator in activities such as course enrolment, course migration, general training on the use of e learning platform to ensure that the courses are well blended with e-learning.



First Year Students of SCIT E-Learning Training on 14th August 2024 in Lab 4 Block

3.1 Undergraduate Students Win Africa 2024 Market place Simulation Award

Our Undergraduate students emerged as winners in the 2024 Marketplace Simulations Awards during the AFRIC2024 Conference in Alexandria, Egypt, held from November 4-15, 2024. The team, comprising fourth-year Software Engineering students Fortunate Tusiime Agnes and Nampwera Elisha, alongside third-year Computer Science student Atuhaire Ronnie, excelled in a rigorous competition featuring participants with advanced degrees, including master's and Ph.D. holders. Guided by Dr. Swaibu Kyanda Kaawaase, Head of the Department of Networks, the team's victory underscores Makerere University's commitment to fostering innovation and entrepreneurial excellence.

Their winning project, Kakati Fit Bikes, emphasized climate action through a simulated business environment. "Kakati," derived from Luganda, translates to "now," symbolizing urgency in addressing climate challenges. This accomplishment earned the team a fully-funded trip to the United States next year for the UN Convention on Climate Change.

Participants utilized a simulation tool widely recognized in institutions like Oxford and MIT to model business environments, competing against teams from Kenya, Niger, Egypt, and Ghana. The event, organized by Marketplace and the University of Senghor, was led by Prof. Dr. Mahamadou Biga-Diambedou. Sponsorship came from Prof. Tonny Oyana, who funded airfare, while accommodation and meals were provided by Prof. Dr. Ernest R. Cadotte.



3.2 Leveraging Artificial Intelligence for Disease Screening

The Ocular project has made significant progress in developing innovative tools for disease diagnostics, focusing on malaria, tuberculosis, and cervical cancer. These products combine advanced technology with practical healthcare solutions to enhance diagnostic accuracy and efficiency in resource-limited settings.



Conference Organisers, Sponsors and the Makerere University Team

3.2.1 Adapter Design and Fabrication

The project successfully designed and fabricated 3D-printed adapters that can be mounted on microscopes used in diagnosing malaria, tuberculosis, and cervical cancer. These adapters were field-tested in health centers and refined based on feedback to ensure optimal performance. Documentation of experiments and surveys conducted during the modification process is currently underway to guide future improvements.

Adapter designs
fabricated in Mak
AI-Health Lab



3.2.2 Malaria Diagnosis App

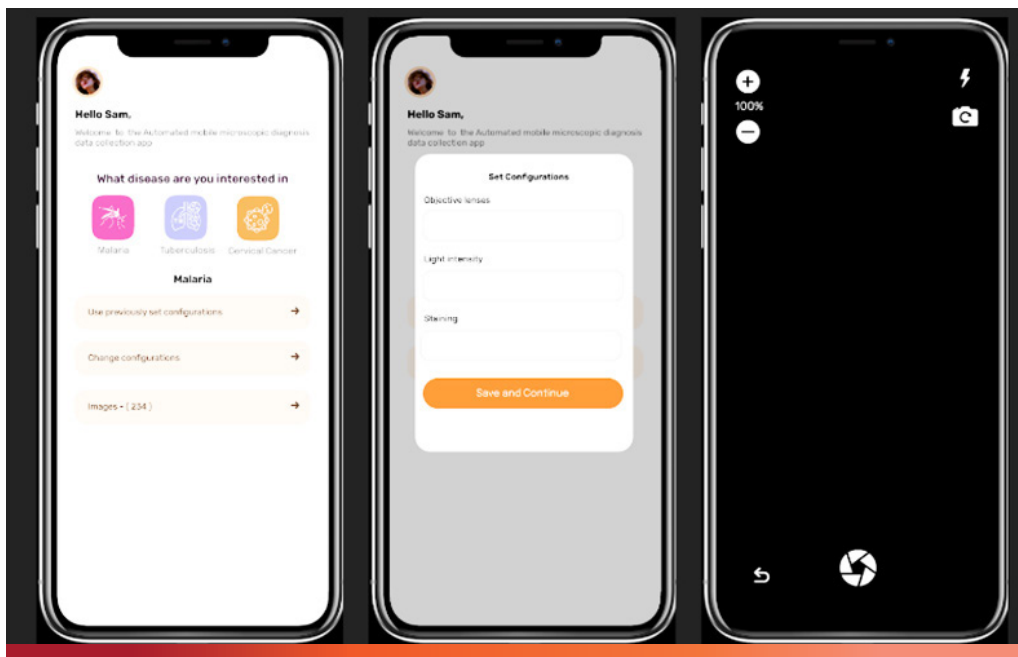
An Android app was created using AI-driven object detection technology to improve malaria diagnosis. The app allows users to capture or upload microscopy images, which the AI system processes to detect and highlight white blood cells and malaria trophozoites using bounding boxes. This tool improves diagnostic accuracy, making it especially useful for healthcare professionals working in resource-limited environments.



Testing of Adapters and data collection app for microscopy data collection

3.2.3 Data Collection App (Malaria and Cervical Cancer)

This app facilitates the systematic collection of microscopy data related to malaria and cervical cancer. It captures images along with important metadata, such as blood slide details and microscope settings. The data is uploaded to a central server, contributing to the creation of a robust dataset that can be used for training AI models and refining diagnostic tools.



Data collection application for malaria, TB and Cervical Cancer Images

3.2.4 Microscopy Teaching Aid

A desktop and mobile app designed to enhance microscopy education was also developed. Students and professionals can capture images, which are then uploaded with metadata for annotation and instructional feedback. This interactive platform improves diagnostic skills by providing hands-on learning experiences, making it a valuable educational tool for both students and professionals. The project work aims to integrate these AI-enhanced tools into the existing health care infrastructure, making disease surveillance and diagnosis more accessible and reliable in Uganda and potentially other similar contexts.

3.3 Air quality Monitoring Devices for Africa

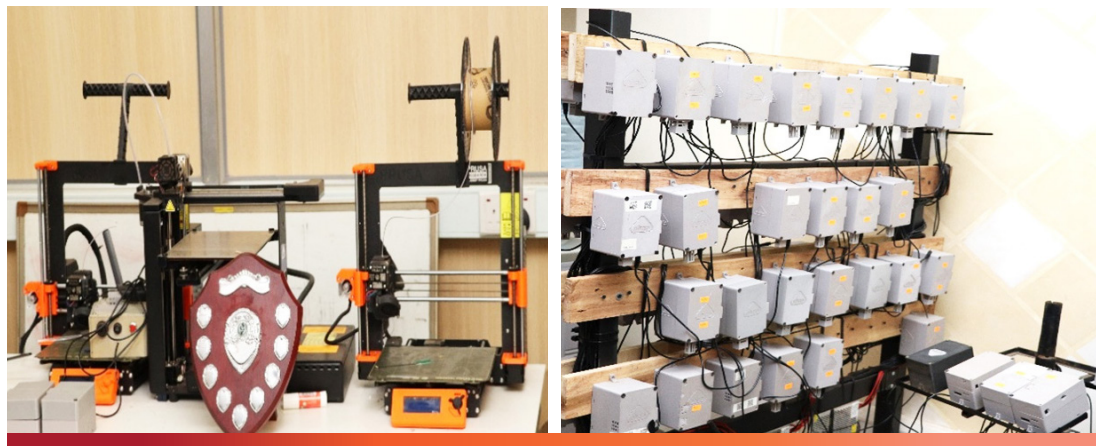
AirQo was founded in 2015 at Makerere University College of Computing and Information Sciences to close the gaps in air quality monitoring across Africa. The current low-cost air quality monitors are designed to suit the African infrastructure, providing locally-led solutions to African air pollution challenges. 2024 was defined by remarkable growth at AirQo, we led the way in advancing local and regional collaborations to improve air quality across African cities.



AirQo device installation in Kisumu Kenya and Kampala Uganda



AirQo has expanded the monitoring network to more than 12 African cities providing actionable data and information on air pollution affecting over 60 million people living in these cities. With the support of our partners, we continue to leverage AI and low-cost sensors to help African cities have access to local air quality data and information to protect human health. Through these efforts, we are enhancing monitoring capacity in major cities across Africa while democratizing access to local air quality data through our AI-powered platforms, thereby empowering individuals, policymakers, and researchers to take data-driven actions to manage air pollution.



Some of the air monitoring devices made in the Mak AirQo lab

3.3.1 AirQo second edition of the CLEAN-Air Forum in Lagos, Nigeria

AirQo held the second edition of the CLEAN-Air Forum in Lagos, Nigeria bringing together more than 200 global experts and over 80 policymakers. For five days, the air quality communities of practice including policymakers, researchers, journalists, development partners, and funding agencies convened to discuss collective actions for sustained air quality interventions in African Cities. The Forum, held under the theme “Advancing collaborations and multi-regional partnerships for clean air actions in African cities,” aimed to promote transdisciplinary collaborations and partnerships in tackling urban air pollution.

3.3.2 Deputy Assistant Secretary for West Africa in the US. Department of States Bureau of African Affairs H.E Will Stevens Visits AirQo’s Air Quality Monitoring Efforts in Nairobi, Kenya

The college was also honored to welcome H.E Will Stevens, Deputy Assistant Secretary for West Africa in the US. Department of States Bureau of African Affairs to witness our ongoing efforts in enabling air quality data access in Nairobi, Kenya. In collaboration with the Nairobi City County Government and World Resources Institute Africa, AirQo showcased joint efforts in Nairobi including a collocation reference site and how they are using motorcycles (locally known as boda boda) to map the city's air quality.



H.E Will Stevens, Deputy Assistant Secretary for West Africa in the US. Department of States Bureau of African Affairs (3rd) and Assoc. Prof. Engineer Bainomugisha (4th) during the visit to AirQo's Air Quality Monitoring Efforts in Nairobi, Kenya

3.3.3 IoT-ra Lab Secures Copyrights for Four Innovative Software Solutions

Makerere University's Internet of Things Research and Application (IoT-ra) Lab secured copyrights for four innovative software solutions developed under the WIMEA-ICT project marking a significant milestone in the lab's technological advancement efforts. The software covering weather information dissemination, data repositories, AWS condition monitoring, and AWS firmware was officially registered with the Uganda Registration Services Bureau. Under the leadership of Dr. Julianne Sansa-Otim, the lab's solutions have been deployed across Uganda and other African countries, contributing to improved weather monitoring and data management in the region.

3.4 Publications

CoCIS remains at the forefront of research and knowledge dissemination, contributing significantly to advancements in computing and information sciences. Through the underlisted high-impact publications in reputable journals and conferences, faculty and students continue to push the boundaries of innovation in areas such as artificial intelligence, data science, cybersecurity, human-computer interaction, and software engineering.

1. Abong, K and Luyombya, D. (2024). Management Practices and Service

- Delivery: A Case of Academic Registrar's Department at the All-Saints University Lango. *Proceedings of the 25th Standing Conference of Eastern, Central and Southern African Library and Information Associations (SCECSAL XXV) held on 22nd – 26th April 2024 in Mombasa, Kenya*. PP5-14.
2. Namujuzi, S and Luyombya, D. (2024). Frameworks for Enhancing Access to Information Resources in Public Institutions in Uganda. *Proceedings of the 25th Standing Conference of Eastern, Central and Southern African Library and Information Associations (SCECSAL XXV) held on 22nd – 26th April 2024 in Mombasa, Kenya*. PP.312 -321.
 3. Baguma, R. (2024, October). Towards Integration and Data Sharing to Enhance Child Protection in Uganda. *In Proceedings of the 17th International Conference on Theory and Practice of Electronic Governance* (pp. 414-424).
 4. Badru, S., Nakibuule, R., Nakatumba-Nabende, J. and Marvin, G., 2024, May. Local Interpretable Model-Agnostic Explanations for Ensemble Learning to Predict Viral Rebound in Patients on Antiretroviral Therapy. *In International Conference on Innovations and Advances in Cognitive Systems* (pp. 393-411). Cham: Springer Nature Switzerland.
 5. Baguma¹, R., 2024. Check for Examining Potential Harms of Large Language Models (LLMs) in Africa Rehema Baguma¹), Hajarah Namuwaya¹, Joyce Nakatumba-Nabende¹, and Qazi Mamunur Rashid². *In Safe, Secure, Ethical, Responsible Technologies and Emerging Applications: First EAI International Conference, SAFER-TEA 2023, Yaoundé, Cameroon, October 25-27, 2023: Proceedings* (Vol. 566, p. 3). Springer Nature.
 6. Bagyendera, M., Peter, N., Brian, G., & Josephine, N. (2024). Contextualizing Syntactic Interoperability Data Standards for Health Information Exchange in Uganda's Public Healthcare System. *In 10th International Conference on Information and Communication Technologies for Ageing Well and e-Health, ICT4AWE 2024* (pp. 267-275). Science and Technology Publications, Lda.
 7. Ibrahim, T., Isaac, K. B., Francis, B., Lule, E., Hellen, N., Chongomweru, H., & Marvin, G. (2024, July). Interpretable Machine Learning Techniques for Predictive Cattle Behavior Monitoring. *In 2024 2nd International Conference on Sustainable Computing and Smart Systems (ICSCSS)* (pp. 1219-1224). IEEE.
 8. Moses, B., Peter, N., Brian, G. and Josephine, N., 2024. Contextualizing Syntactic Interoperability Data Standards for Health Information Exchange in Uganda's Public Healthcare System. *In 10th International Conference on Information and Communication Technologies for Ageing Well and e-Health, ICT4AWE 2024* (pp. 267-275). Science and Technology Publications, Lda.
 9. Mpirirwe, H., Mirembe, D.P., Lubega, J.T., Agaba, J. and Nanyonga Clarke, R., 2024, January. A Multifactor Authentication Framework for Usability in Education Sectors in Uganda. *In International Conference on Data Management, Analytics & Innovation* (pp. 205-216). Singapore: Springer

- Nature Singapore.
10. Mugabi, S., Onen, S., Marvin, G., & Ndibatya, I. (2024). Leveraging Visual Analytics to Explore Bike Imbalance in Localised Operational Environments. In *International Conference on Communication and Computational Technologies* (pp. 461-475). Singapore: Springer Nature Singapore.
 11. Nalule, D., Edekebon, E., Nagwovuma, M., Jjingo, D., & Marvin, G. (2024). Explainable Deep Learning Approaches to Credit Risk Evaluation. In *Proceedings of the 2024 Sixteenth International Conference on Contemporary Computing* (pp. 198-206).
 12. Akello, C. K., & Nabukenya, J. (2024). UCD–CE Integration: A Hybrid Approach to Reinforcing User Involvement in Systems Requirements Elicitation and Analysis Tasks. *Group Decision and Negotiation*, 33(6), 1355-1403.
 13. Akello, C. K., & Nabukenya, J. (2024). Users involvement in the electronic health information systems development process in Uganda: what is missing in relation to requirements gathering and analysis. *Oxford Open Digital Health*, 2, oqae020.
 14. Alunyu Egwar, A., Amiyo, M. R., & Nabukenya, J. (2024). Framework for standardizing digital health in resource-constrained settings: a case study of Uganda's digital health communication infrastructure. *Oxford Open Digital Health*, 2, oqae018.
 15. Alunyu Egwar, A., Amiyo, M.R. and Nabukenya, J., 2024. Framework for standardizing digital health in resource-constrained settings: a case study of Uganda's digital health communication infrastructure. *Oxford Open Digital Health*, 2, p.oqae018.
 16. Alunyu, A. E., Amiyo, M. R., & Nabukenya, J. (2024). Contextualised digital health communication infrastructure standards for resource-constrained settings: Perception of digital health stakeholders regarding suitability for Uganda's health system. *PLOS Digital Health*, 3(9), e0000603.
 17. Azamuke, D., Katarahweire, M. and Bainomugisha, E., 2024. MoMTSim: A Multi-agent-based Simulation Platform Calibrated for Mobile Money Transactions. *IEEE Access*. 한국통신학회 인공지능 학술대회 논문집
 18. Bainomugisha, E. and Nakatumba-Nabende, J. (2024), Developing and Deploying End-to-End Machine Learning Systems for Social Impact: A Rubric and Practical Artificial Intelligence Case Studies From African Contexts. *Applied AI Letters* e100. <https://doi.org/10.1002/ail2.100>
 19. Bainomugisha, E., Warigo, P.A., Daka, F.B., Nshimye, A., Birungi, M. and Okure, D., 2024. AI-driven environmental sensor networks and digital platforms for urban air pollution monitoring and modelling. *Societal Impacts*, 3, p.100044.
 20. Bainomugisha, Engineer, et al. "Computer Science Education in Selected Countries from Sub-Saharan Africa." *ACM Inroads* 15.1 (2024): Pages 64 – 82 <https://doi.org/10.1145/3643037>.

21. Becker, G., Namanya, P., Kiganda, C., Nabukenya, J., Wendt, L., Rukundo, G., ... & Etima, J. (2024). HIV Knowledge and Sexual Behaviors in Perinatally Infected Ugandan Youth: A Cross-Sectional Survey. *Journal of the International Association of Providers of AIDS Care (JIAPAC)*, 23, 23259582241299712.
22. Bhutada, K., Venkateswaran, M., Atim, M., Munabi-Babigumira, S., Nankabirwa, V., Namagembe, F., ... & Papadopoulou, E. (2024). Factors influencing the uptake of antenatal care in Uganda: a mixed methods systematic review. *BMC pregnancy and childbirth*, 24(1), 730.
23. Bitwire, G. A., & Han, D. S. (2024). Benchmarking YOLOv8, YOLOv9, and YOLOv10 for Detecting Tiny and Small Objects in Complex Environments. , 4
24. Eilu, E., Pettersson, J. S., Baguma, R., & Bhutkar, G. (2024). Rethinking ICT Adoption Theories in the Developing World: Information and Communication Technologies. Springer Nature
25. Eilu, E., Pettersson, J.S., Baguma, R. and Bhutkar, G., 2024. Rethinking ICT Adoption Theories in the Developing World: Information and Communication Technologies.
26. Eun, S. W., Bitwire, G. A., Thu, N. T. H., Kang, S. J., & Han, D. S. (2024, October). Non-Calibration Sensor Fusion Using High Resolution 3D LiDAR. In *2024 15th International Conference on Information and Communication Technology Convergence (ICTC)* (pp. 158-159). IEEE.
27. Ferretti, A., Adjei, K.K., Ali, J., Atuire, C., Ayuk, B.T., Banounin, B.H., Cengiz, N., Gichoya, J., Jjingo, D., Juma, D.O. and Kotze, W., 2024. Digital tools for youth health promotion: principles, policies and practices in sub-Saharan Africa. *Health promotion international*, 39(2), p.daae030.
28. Gahwera, T.A., Eyobu, O.S. and Isaac, M., 2024. Analysis of Machine Learning Algorithms for Prediction of Short-term Rainfall Amounts Using Uganda's Lake Victoria Basin Weather Dataset. *IEEE Access*.
29. Guma, I. P., Rwashana, A. S., Oyo, B., & Waiswa, D. (2024). Modeling Land-Use Land-Cover Change Complexity: A System Dynamics Approach. *International Journal of System Dynamics Applications (IJSDA)*, 12(1), 1-22.
30. Joseph, T., Kenneth, M. H., & Rose, N. (2024). Explainable AI for Transparent and Trustworthy Tuberculosis Diagnosis: From Mere Pixels to Actionable Insights. *East African Journal of Information Technology*, 7(1), 341-354.
31. Kanagwa, B., Kahiigi, E.K., Semwanga, A.R., Siminyu, S. and Lubowa, P., 2024. Towards Gamification of Science and Mathematics in Upper Primary Schools in Uganda.
32. Kanyerezi, S., Nabisubi, P., Kebirungi, G., Sserwadda, I., R. Kidenya, B., Jjingo, D. and Mboowa, G., 2024. Metagenomics insights into the microbial resistome and virulome composition of Kampala's wastewater. *Open Research Africa*, 7, p.8.

33. Kapalaga, G., Kivunike, F. N., Kerfua, S., Jjingo, D., Biryomumaisho, S., Rutaisire, J., ... & Kiwala, Y. (2024). A unified foot and mouth disease dataset for Uganda: evaluating machine learning predictive performance degradation under varying distributions. *Frontiers in Artificial Intelligence*, 7, 1446368.
34. Kenneth, M. H., Joseph, T., & Rose, N. (2024). AI Methods and Algorithms for Diagnosis of Intestinal Parasites: Applications, Challenges and Future Opportunities. *East African Journal of Information Technology*, 7(1), 366-379.
35. Kibet, C.K., Entfellner, J.B.D., Jjingo, D., de Villiers, E.P., de Villiers, S., Wambui, K., Kinyanjui, S. and Masiga, D., 2024. Designing and delivering bioinformatics project-based learning in East Africa. *BMC bioinformatics*, 25(1), p.150.
36. Kobusingye, G. and SSendagi, M., 2024. The Various Types Of Supply Chain Management Systems Used By Coca-Cola Bottling Company In Namanve. *SJ Business Research Africa*, 1(1), pp.15-15.
37. Lilian, K., 2024. A Collaboration Process for Developing Business Process Improvement Implementation Plans. In *Rethinking ICT Adoption Theories in the Developing World: Information and Communication Technologies* (pp. 119-144). Cham: Springer Nature Switzerland.
38. Lutalo, J.W. and Oyana, T., 2024. VOSA: A Reusable and Reconfigurable Voice Operated Support Assistant Chatbot Platform. *Available at SSRN 4810799*.
39. Magumba, M. A., Denton, T., & Bashir, M. (2024). A bird vocalisation dataset of birds in Uganda for automated bio-acoustic monitoring and analysis. *Data in Brief*, 54, 110433.
40. Manuela, U. M., Nakasi, R., Jjingo, D., Hellen, N., Ngobye, M., & Marvin, G. (2024, July). Machine Vision Intelligence Using Layer-Wise Relevance Backward Propagation For Breast Cancer Diagnosis. In *2024 5th International Conference on Image Processing and Capsule Networks (ICIPCN)* (pp. 143-149). IEEE.
41. Mir, M. S., Moses, G., Gulzar, Y., & Reegu, F. A. (2024). The Impact of Interplay between Intrinsic Capabilities, Extrinsic Support and System Quality on e-Learning Service Experiences. *IEEE Access*.
42. Muinda, P.E., Basaza-Ejiri, A.H., Maiga, G. and Mayoka, K., 2024. A Digitization Model in Ugandan Technical, Vocational and Education Training Institutions: An Agency Theory Perspective.
43. Mukalazi, S. H., Jianquan, O., Namatovu, H. K., & Kimbugwe, N. Towards an Integrated Rat Llm Framework for Contextually Appropriate Complex Clinical Recommendations. *Available at SSRN 4668689*.
44. Murindanyi, S., Nakatumba-Nabende, J., Sanya, R., Nakibuule, R. and Katumba, A., 2024. Enhanced Infield Agriculture with Interpretable Machine Learning Approaches for Crop Classification. *arXiv preprint arXiv:2408.12426*.
45. Musiimaami, D., 2024. Information Needs, and Sources of Information, of Art and Design Students in Diploma Awarding Institutions in Uganda. *Library Philosophy and Practice*, pp.1-23.

46. Mwotil, A., Anderson, T., Kanagwa, B., Stavrinos, T. and Bainomugisha, E., 2024. LowPaxos: State Machine Replication for Low Resource Settings. *IEEE Access*.
47. Nagwovuma, M., Maiga, G., & Nakakawa, A. (2024). Discriminant Validity of Factors for Evaluating Performance of eHealth Information Systems. In *Rethinking ICT Adoption Theories in the Developing World: Information and Communication Technologies* (pp. 47-70). Cham: Springer Nature Switzerland.
48. Nakakawa, A., de Manuel Keenoy, E., Zabala, A.F., Berastegui, D.V. and Suarez, J.T., 2024. Strategic objectives for aligning healthcare and IT practices in providing integrated care for multimorbid patients: a soft systems methodology perspective. *Systemic Practice and Action Research*, 37(1), pp.59-80.
49. Nakakawa, A., de Manuel Keenoy, E., Zabala, A.F., Berastegui, D.V. and Suarez, J.T., 2024. Mutual adoption of Soft Systems Methodology, Co-creation, Enterprise Architecture, and Balanced Scorecard for Continuous Assessment and Improvement of Programmes on Integrated Care for Multimorbid Patients. *Systemic Practice and Action Research*, 37(4), pp.351-386.
50. Nakakawa, A., Kasule, D., Kasusse, M., Guma, G., Aisu, S., Ogwok, P. A., & Sendagire, H. (2024). Process for Leveraging Enterprise Architecture in Information Systems Strategic Planning: A Case of Developing a Strategy and Master Plan for a National Integrated Health Laboratory Information Management System in Uganda. *Complex Systems Informatics and Modeling Quarterly*, (40), 58-93.
51. Nakakawa, A., Tulinayo, F.P., Tabo, G. O., Bommel, P.V., Mulder, H. B., & Proper, H. A. (2024). Supplementing the Build Activity in Design Science Research with Soft Systems Methodology: A Technique of Creating Frameworks for Guiding Interventions Against Unstructured Problems.
52. Nakatumba, Nabende, J., Babirye, C., Nabende, P., Tusubira, J. F., Mukiibi, J., Wairagala, E. P., ... & Katumba, A. (2024). Building Text and Speech Benchmark Datasets and Models for Low-Resourced East African Languages: Experiences and Lessons. *Applied AI Letters*, 5(2), e92.
53. Nakatumba-Nabende, J., Mukiibi, J., Bateesa, T.S., Murindanyi, S., Katumba, A. and Mutebi, C., 2024. Machine Learning Analysis of Radio Data to Uncover Community Perceptions on the Ebola Outbreak in Uganda. *ACM Journal on Computing and Sustainable Societies*.
54. Namatovu, H. K., & Arinaitwe, I. (2024). Student adoption of e-learning at makerere university during COVID-19: A structural equation modelling based on the unified theory of acceptance and use of technology. *E-Learning and Digital Media*, 20427530241254537.
55. Namatovu, H.K. and Arinaitwe, I., 2024. Student adoption of e-learning at makerere university during COVID-19: A structural equation modelling

- based on the unified theory of acceptance and use of technology. *E-Learning and Digital Media*, p.20427530241254537.
56. Namugera, L., Bukirwa, J. and Okello-Obura, C., 2024. Examining a new generation of service provision in a developing country: Opportunities and Challenges of Virtual Library Services in Universities in Uganda. *Qualitative and Quantitative Methods in Libraries*, 14(2), pp.195-205.
 57. Nantongo, B. A., Nabukenya, J., Nabende, P., & Kamulegeya, J. (2024). A retrospective cohort study on predicting infants at a risk of defaulting routine immunization in Uganda using machine learning models. *JAMIA open*, 7(4), ooae132.
 58. Nsubuga, M., Galiwango, R., Jjingo, D. and Mboowa, G., 2024. Generalizability of machine learning in predicting antimicrobial resistance in E. coli: a multi-country case study in Africa. *BMC genomics*, 25(1), p.287.
 59. Nsubuga, M., Mutegeki, H., Jjingo, D., Munube, D., Namazzi, R., Opoka, R., Kasirye, P., Ndeezi, G., Hume, H., Mupere, E. and Kebirungi, G., 2024. The Ugandan sickle Pan-African research consortium registry: design, development, and lessons. *BMC medical informatics and decision making*, 24(1), p.212.
 60. Ogenrwot, D., Nakatumba-Nabende, J., Businge, J. and Chaudron, M.R., 2024. Empirical Investigation of the Relationship Between Design Smells and Role Stereotypes. *arXiv preprint arXiv:2406.19254*.
 61. Ongaya, K., Alidri, A., Bagarukayo, E., Oyo, B., Bazibu, C.M. and Luyimbazi, G., 2024. Ontology of Plagiarism: the Non Academic Perspectives. *International Journal of Advanced Research*, 7(1), pp.55-70.
 62. Paul, M., Nagwovuma, M., Nansamba, B., Hellen, N., Jingo, D., & Marvin, G. (2024). Trustworthy Deep Learning Techniques for Credit Risk Assessment. In *2024 5th International Conference on Electronics and Sustainable Communication Systems (ICESC)* (pp. 1949-1962). IEEE.
 63. Please, H., Narang, K., Bolton, W., Nsubuga, M., Luweesi, H., Richards, N.B., Dalton, J., Tendo, C., Khan, M., Jjingo, D. and Bhutta, M.F., 2024. Virtual reality technology for surgical learning: qualitative outcomes of the first virtual reality training course for emergency and essential surgery delivered by a UK–Uganda partnership. *BMJ Open Quality*, 13(1), p.e002477.
 64. Safari, Y., Nakatumba-Nabende, J., Nakasi, R. and Nakibuule, R., 2024. A Review on Automated Detection and Assessment of Fruit Damage Using Machine Learning. *IEEE Access*.
 65. Sanya, R., Nabiryo, A.L., Tusubira, J.F., Murindanyi, S., Katumba, A. and Nakatumba-Nabende, J., 2024. Coffee and cashew nut dataset: A dataset for detection, classification, and yield estimation for machine learning applications. *Data in Brief*, 52, p.109952.
 66. Shaw, J., Ali, J., Atuire, C.A., Cheah, P.Y., Español, A.G., Gichoya, J.W., Hunt, A., Jjingo, D., Littler, K., Paolotti, D. and Vayena, E., 2024. Research ethics

- and artificial intelligence for global health: perspectives from the global forum on bioethics in research. *BMC Medical Ethics*, 25(1), p.46.
67. Soomro, S., Bano, A., Gulzar, Y., & Moses, G. (2024). Factors measuring the readiness of teacher educators to use web 2.0 technologies. *Discover Sustainability*, 5(1), 501.
68. Ssekagiri, A., Jjingo, D., Bbosa, N., Bugembe, D.L., Kateete, D.P., Jordan, I.K., Kaleebu, P. and Ssemwanga, D., 2024. HIVseqDB: a portable resource for NGS and sample metadata integration for HIV-1 drug resistance analysis. *Bioinformatics Advances*, 4(1), p. vbae008.
69. Toure, E.N., Victor Nthusi, V., Garland, R., Mbandi, A., Bainomugisha, E., Hodoli, C.G., Doumbia, M., Mbiake, R., Klopper, D., Ojeh, V.N. and Gatari, M., 2024, August. Opportunities for Promoting Integrated Action on Air Pollution and Climate in Africa. In *ISEE Conference Abstracts* (Vol. 2024, No. 1).
70. Wright, C.Y., Jaca, A., Kapwata, T., Naidoo, N., Awokola, B., Bainomugisha, E., Berhane, K.T., Blesic, S., Kehbila, A., Naidoo, R. and Nemukula, B., 2024. Identifying globally relevant learnings from Africa's challenges and solutions to climate change and air pollution-related health impacts: a data science scoping review protocol. *BMJ open*, 14(5), p.e076941.
71. Zhang, W. and Kobusingye, G., 2024. Comparing the Effects of Social Media Influencers and Electronic Word-of-Mouth on Consumer Purchase Intention for Outdoor Branded Products in China.
72. Musiimaami, D.; Okello-Obura, C and Luyombya, D. (2024). Information-seeking Strategies Used and Challenges Encountered by Art and Design Students in Uganda When Seeking Information. *The Uganda Higher Education Review Journal* 11(1), 79-93. DOI: <https://doi.org/10.58653/nche.v11i1.07>
73. Musiimaami, D.; Okello-Obura, C and Luyombya, D. (2024). Information-seeking Strategies Used and Challenges Encountered by Art and Design Students in Uganda When Seeking Information. *The Uganda Higher Education Review Journal* 11(1), 79-93. DOI: <https://doi.org/10.58653/nche.v11i1.07>
74. Luyombya, D. (2024). "Archiving, Memory and Method from the Global South". A keynote address on Archiving, Memory and Methods from the Global South, College of Humanities and Social Sciences (CHUSS), Makerere University, 17–19 October 2024.
75. Namujuzi, S and Luyombya, D. (2024). Frameworks for Enhancing Access to Information Resources in Public Institutions in Uganda. *Proceedings of the 25th Standing Conference of Eastern, Central and Southern African Library and Information Associations (SCECSAL XXV) held on 22nd – 26th April 2024 in Mombasa, Kenya*. PP.312 -321.
76. Kaddu, S., Ssekitto, F., Nalunkuma, E., & Masenya, T. M. (2024). Beyond the stacks: Reimagining collection development at the Uganda institute of information and communication technology library to champion SDG 16. *Information Development*, 02666669241299038.

77. Masenya, T. M., Ssekitto, F., Kaddu, S., & Simati, S. (2024). Management of Electronic Health Records in Virtual Health Environments: The Case of Rocket Health in Uganda. *International Journal of Healthcare Information Systems and Informatics (IJHISI)*, 19(1), 1-17.
78. Mary Nalumansi and Sarah Kaddu (2024). Cultivating a lifelong learning for reading: a concept framework for enhancing Children's Literacy through framework for enhancing children's literacy through Community Libraries in Uganda Ghana Library journal-2024, Vol.29(2)
79. Namugera, L., Bukirwa, J., & Okello-Obura, C. (2024). Examining a new generation of service provision in a developing country: Opportunities and Challenges of Virtual Library Services in Universities in Uganda. *Qualitative and Quantitative Methods in Libraries*, 14(2), 195-205. Available at <https://www.qqml-journal.net/index.php/qqml/article/view/842/748>
80. Okello, G.; Bukirwa, J & Magara, E. Exploring Control of Access to the Japadhola Indigenous Information. *IFLA Journal* (Accepted for publication)
81. Nabutto, C.C.L., Mutibwa, L. N., Mwanje A. & Kamooga N. (2024). Integration of Artificial Intelligence Tools in Libraries 2012-2023: A Bibliometric Assessment. *IJFMR* Volume 6, Issue 4, July-August 2024. DOI 10.36948/ijfmr.2024.v06i04.25168
82. Musiimaami, D.; Okello-Obura, C and Luyombya, D. (2024). Information Needs, and Sources of Information, of Art and Design Students in Diploma Awarding Institutions in Uganda *Library Philosophy and Practice*, 1-23. At: <https://www.proquest.com/scholarly-journals/information-needs-sources-art-design-students/docview/3061354843/se-2?accountid=169196>
83. Musiimaami, D.; Okello-Obura, C and Luyombya, D. (2024). Information-seeking Strategies Used and Challenges Encountered by Art and Design Students in Uganda When Seeking Information. *The Uganda Higher Education Review Journal* 11(1), 79-93. DOI: <https://doi.org/10.58653/nche.v11i1.07>
84. Bukenya L., Eyobu O.S., and Oyana T.J., "An LSTM Model for Diagnosis Prediction of Lower Respiratory Tract Infections using a Minimalistic Data set," in *IEEE Access*, doi: 10.1109/ACCESS.2025.3540359.
85. Eriksson G, Kamwesiga J, Fors U, Oyana T, von Koch L, Ytterberg C, Guidetti S. A randomized controlled trial of mobile phone-supported and family-centred rehabilitation after stroke in Uganda, F@ce 2.0: a study protocol. *JMIR Research Protocols* 2024;0:e0. <https://www.researchprotocols.org/2024/0/e0/>. doi: 10.2196/60955.
86. Namatovu, HK., Magumba MA., Oyana TJ. 2023. A Water Quality Dataset of Levels of Metal, Nutrient and Anions in Sample Water Points from Sixteen Selected Urban and Rural Districts of Uganda. *Data in Brief*. 109601(50): 2352-3409. <https://doi.org/10.1016/j.dib.2023.109601>.
87. Tumusiime, AG., Odongo, SE., Mugume, I., Oyana, TJ. 2023. A Weather Features Dataset for Prediction of Short-term Rainfall Quantities in Uganda. *Data in Brief*. 109613(50): 2352–3409. <https://doi.org/10.1016/j.dib.2023.109613>.

3.5 Research Grants

CoCIS has successfully secured multiple research grants through competitive proposal writing. These grants not only supplement the university's resource envelope but also enhance human resource capacity and support cutting-edge research. The table 4 below provides a detailed list of the awarded grants.

Table 4 List of Grants Received for Research

SN	Name of Project	Funder & Country	PI Name	Project Period (Start - End)	Currency	Budget Amount
1	Managing Hub for Artificial Intelligence for Sexual and Reproductive Health in Africa: AI4MSRH Hub	IDRC (Canada)	Assoc. Prof. Engineer Bainomugisha	1st Nov 2021 - 31st Oct 2024	USD	137,867
2	Clean Air for Healthier Cities: Scaling up Participatory and Collaborative Monitoring, Modelling, and Advocacy for Air Quality Management	US Department of State	Assoc. Prof. Engineer Bainomugisha	30th Sept 2022 - 30th Sept 2025	USD	478,835.60
3	Scaling up AirQo to African Cities	Google (USA)	Assoc. Prof. Engineer Bainomugisha	23rd Nov 2021 - 31st July 2025	USD	3,000,000
4	Crane Cloud	Paul Maritz (USA)	Assoc. Prof. Engineer Bainomugisha	25th June 2018	USD	306,920.24
5	Establishing DeepMind AI Scholarships at Makerere University	DeepMind Technologies (UK)	Assoc. Prof. Engineer Bainomugisha	May 2023 - 2024	USD	93,980
6	Google Research Awards	Google LLC (USA)	Assoc. Prof. Engineer Bainomugisha		USD	100,000

7	AirQount: Autonomous Vehicle Emissions Estimation for Low-Resource Cities	Arm Limited (UK)	Assoc. Prof. Engineer Bainomugisha	25th June 2024 - 30th June 2025	USD	249,900
8	Leveraging Data Science Applications to Improve Children's Environmental Health in Sub-Saharan Africa (DICE)	NIH	Assoc. Prof. Engineer Bainomugisha	Sept 2023 - Aug 2026	USD	98,508
9	Benchmarking Automatic Speech Recognition Models for African Languages	Bill & Melinda Gates Foundation	Dr. Joyce Nakatumba Nabende	21st March 2024 - 31st March 2025	USD	249,400
10	TWAS-UNESCO (SG-NAPI)	UNESCO	Dr. Joyce Nakatumba Nabende		USD	9,100
11	RTB Breeding: A Consolidated Environment	International Potato Center (Peru)	Dr. Joyce Nakatumba Nabende	16th Dec 2023 - 31st Dec 2024	USD	56,923
12	RTB Cassava Breeding	International Potato Center (Peru)	Dr. Joyce Nakatumba Nabende	1st Jan 2023 - 31st Dec 2024	USD	40,000
13	Applied Human- Machine Intelligence in East Africa (AHUMAIN)	European Union / Erasmus Program	Dr. Joyce Nakatumba Nabende	1st Jan 2023 - 31st Dec 2025	USD	141,142
14	Understanding Gender Biases in Building Automatic Speech Recognition Systems in Africa (GRAIN)	IPAR (Senegal)	Dr. Joyce Nakatumba Nabende	1st Aug 2023 - 30th Nov 2024	USD	41,080
15	Project Amplify - Improving AI through Community-Based Feedback	Google LLC	Dr. Joyce Nakatumba Nabende	Sept 2023 - July 2024	USD	200,000

16	Extrapolating Counts from Open Buildings with Sentinel-2	Google Research Collabs Africa	Sanya Rahman	-	USD	35,000
17	Analyzing ChatGPT for Cross-Lingual Agricultural Advisory for Smallholder Farmers	Bill & Melinda Gates Foundation	Dr. Joyce Nakatumba Nabende	10th July 2023 - 30th April 2024	USD	150,000
18	Waxal Project Partnership (Google Voices)	Google LLC	Dr. Joyce Nakatumba Nabende	July 2023 - April 2024	USD	345,000
19	OCULAR: AI Automated Mobile Microscopic Diagnosis of Malaria, Cancer & Tuberculosis	Google (USA)	Dr. Rose Nakasi	Sept 2023 - 2026	USD	1,500,000
20	Pathways to Change: Gender Justice in STEM Research (GeJuSTA)	University of Cape Town (South Africa)	Ms. Ssozi Fiona Catherine	1st Jan 2022	ZAR	178,046
21	ICT Labs for the Pathogen Economy	Government of Uganda	Dr. Joyce Nakatumba Nabende	1st June 2022 - 30th Nov 2023	ZAR	1,000,000,000
22	AFRETECH	Carnegie Mellon University Africa	Assoc. Prof. Engineer Bainomugisha	1st Aug 2023 - 31st May 2026	USD	79,695
23	Epilepsy	Makerere University College of Health Sciences	Dr. Daudi Jjingo	1st July 2023 - Dec 2024	USD	11,000
24	Adaptive Environmental Monitoring Network for East Africa (AdEMNEA)	NORHED II	Dr. Julianne Sansa Otim	1st July 2022 - 30th June 2026	NOK	9,359,490
25	Next Generation-Cassava	Bill & Melinda Gates Foundation	Dr. Joyce Nakatumba Nabende	April 2018 - March 2023	USD	334,634

26	Improving Machine Translation for Informal Usage of Ugandan Languages	Google LLC (USA)	Dr. Peter Nabende	1st Nov 2023 - 30th Oct 2024	USD	30,000
27	RIF	-	Dr. Namatovu Hasifah	2023 - 2024	UGX	493,157,714
28	Enhancing Groundnuts Value Chain Information Management Using SUFACE Model	RIF	Dr. Drake Patrick Mirembe	2023 - 2024	UGX	169,622,000
29	Data Science Without Borders	APHRC (Kenya)	Dr. Joyce Nakatumba Nabende	1st Aug 2024 - 31st Dec 2026	USD	336,186
30	RTB Breeding: Consolidated Investment (Cassava)	Bill & Melinda Gates Foundation	Dr. Joyce Nakatumba Nabende	1st Jan 2024 - March 2025	USD	85,386
31	RTB Breeding: Consolidated Investment (Sweet Potatoes)	Bill & Melinda Gates Foundation	Dr. Joyce Nakatumba Nabende	1st Jan 2024 - March 2025	USD	48,100
32	RTB Breeding: Consolidated Investment (NARO)	National Agricultural Research Organisation	Dr. Joyce Nakatumba Nabende	2nd Oct 2024 - 31st March 2025	USD	19,900
33	Makerere University AI for Development Research Lab	IDRC (Canada) & FDCO	Dr. Joyce Nakatumba Nabende	1st Oct 2024 - 30th Nov 2026	USD	574,900
34	AI4D - FDCO Innovation: Scaling AI for Health	IDI	Assoc. Prof. Engineer Bainomugisha	1st Oct 2024 - 30th Sept 2026	USD	100,984

4.0

**COMMUNITY OUTREACH,
KNOWLEDGE TRANSFER
AND PARTNERSHIPS**

The college held over 20 events aimed at knowledge transfer and partnerships including collaborative efforts, workshops and conferences that promote academic and professional growth. These activities also strengthen research capacity and tackle critical societal challenges.

4.1 Community Outreach**4.1.1 UCC and Mak- CoCIS Train 2,643 in Rural Areas to Boost Digital Literacy**

The Uganda Communications Commission (UCC), in collaboration with Makerere University's College of Computing and Information Sciences (CoCIS), conducted a digital skills training program in 15 rural districts, benefiting 2,643 participants from April 1 to June 19, 2024. Funded by the Uganda Communications Universal Service Access Fund (UCUSAF), the program covered ICT essentials such as internet literacy, e-government services, and online safety. Despite positive feedback, challenges included device shortages and malfunctioning smartphones, particularly in Kotido and Bundibugyo. Participants, especially women in seven districts, found the training valuable for economic activities, such as market price comparisons. Recommendations for improvement include addressing network issues, follow-up training, phone repair programs, and providing solar-powered devices, aligning with Uganda's National Development Plan III to enhance digital inclusion.

One of the
beneficiaries
testified



"Fostering Academic Resilience and Long-Term Achievement"

4.1.2 Mak Ocular built capacity for laboratory technicians in health facilities

The Ocular project focuses on enhancing AI-powered diagnostics for malaria, tuberculosis, and cervical cancer through capacity-building initiatives and data collection efforts aimed at healthcare professionals. By training lab technicians and deploying data collection systems across various health facilities, including district hospitals, Health Centre IVs, and Health Centre IIIs, the project aims to improve diagnostic accuracy and patient care. Launched in June 2024, the project gathered critical data from key referral hospitals like Mulago, Kiruddu, Arua, Amudat, and Kambuga, training over 150 technicians and pathologists in using the Ocular mobile app and 3D-printed adapters. Strict protocols for data collection and annotation were implemented to ensure consistency and reliability in AI model training, with cervical cancer images categorized into diagnostic classes and malaria images annotated based on species and growth stages. Feedback from healthcare professionals has played a vital role in refining the technology, ensuring its effective application in real-world healthcare settings.

Pictorial



Lab technicians at Mulago hospital being trained by Solomon Sumba

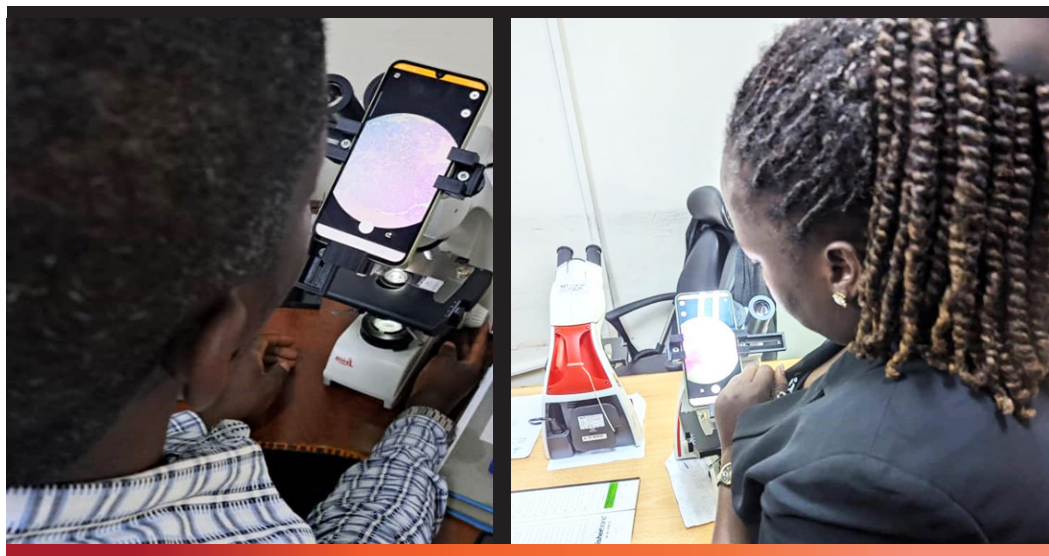
**Mr. Tusubira Jeremy
training Mulago Hospital
lab technicians**



**The technical team received in Kiruddu National Referral Hospital to train
Lab technicians on how data is to be collected.**



PI Dr. Rose Nakasi (2nd back row) witnessing the training of lab technicians and data collection at Uganda Cancer Institute



Data collection at Uganda Cancer Institute

4.2 Workshops, seminars and conferences

4.2.1 Workshop on AI for students and staff participating in the Erasmus+ e program.

The College hosted a two-day workshop on Artificial Intelligence (AI) for students and staff in the Erasmus+ exchange program, starting on November 11, 2024. Participants from Makerere University, Busitema University, and the University of Twente engaged in discussions on AI applications in agriculture and health. Professor Tonny Oyana, Principal of CoCIS, opened the workshop, highlighting AI's growing role across disciplines and urging innovation to stay competitive. Coordinated by Dr. Rose Nakibuule, the program promotes international research collaboration and knowledge exchange between the participating universities.

4.2.2 Workshop on Integration of AI in Library and Information Sciences Education

The East African School of Library and Information Sciences held a workshop about Integration of Artificial Intelligence to enhance Library and Information Sciences Education on July 18th 2024. During this workshop, Teaching staff at EASLIS were trained and updated on various issues which would improve teaching and learning. They explored various applications of AI and developed strategies for integrating AI related topics into existing Library and Information Sciences Courses.

Integrating artificial intelligence (AI) into Library and Information Sciences (LIS) education is becoming increasingly important due to the profound impact AI has on information management, retrieval, and user experience. At CoCIS we believe that these kinds of trainings will enable us stay relevant in the Digital Age and suit the rapidly evolving landscape of Libraries and Information Services. We are certain that future librarians produced from our college will remain relevant and equipped with the skills necessary to thrive in a digital environment.



AI workshop Training Session ongoing



Staff Members pose for a group photo with the College Principal after the workshop.

4.2.3 Pedagogy Workshop

We also held a two-day Library & Information Science (LIS), and Records and Archives Management (RAM) Teaching Professionals Pedagogical Re-Oriented Workshop on June 18th – 19th 2024. The event held at Makerere University College of Computing and Information Sciences (Mak COCIS), Block B attracted a number of teaching staff at various ranks from Ugandan universities, and the region at large. This workshop aimed at improving teaching and learning.



Prof. Tony Oyana, CoCIS Principal giving his remarks

4.2.4 National celebration of D.E.A.R. Day

On March 15, 2024, staff and students from the East African School of Library and Information Sciences (EASLIS) joined the national celebration of D.E.A.R. (Drop Everything and Read) Day, which promotes a positive reading culture in Uganda. EASLIS organized a reading session at 11:00 AM, led by Dr. Peter Nabende, Deputy Principal of the College of Computing and Information Sciences. The event was attended by students, Dean Dr. Sarah Kaddu, the school librarian, and teaching staff. Participants engaged in a silent reading session, followed by student reflections on their experience.

4.2.5 Makerere AI Health Lab and Ocular dissemination activities

The Makerere AI Health Lab and Ocular project have actively engaged in numerous research dissemination activities, showcasing their AI-powered diagnostic innovations at key national and international events. At the national level, the project held an inaugural AI in African Health Conference on November 28, 2024, brought together key stakeholders to discuss AI's role in African health systems. The team also participated in the Annual Media Convention, National Science Week, the UG Dev Summit, and the Health Innovation Conference, demonstrating their AI-powered mobile microscopy tool for disease diagnosis. They engaged policymakers at the Conference of Speakers and Presiding Officers of the

Commonwealth (CSPOC), the IANPHI Conference, and the Uganda National TB and Leprosy Science Summit, advocating for AI's potential in tackling public health challenges. Additionally, Ocular showcased its work at World Malaria Day Colloquium, Data Science Africa Conference, and the ITU GSR-24 conference, emphasizing AI's role in disease diagnosis and public health policy.



PS. Dr. Aminah Zawedde flanked by Dr. Peter Nabende arrives for the Inaugural AI Health conference



Mak staff posing for a group photo with the Guest speakers after the opening session at the Inaugural AI Health conference

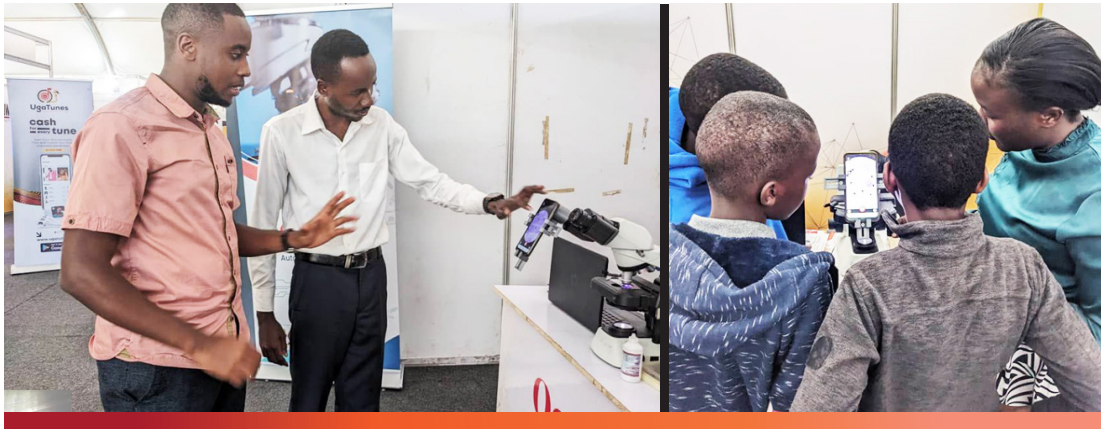
Internationally, the team showcased their innovations at major forums such as the the UNPF ICPD30 Global Dialogue on Technology in New York, the RELX SDG Inspiration Day 2024, and the 10th Annual Conference of the Economic Policy Research Network in Rwanda to discuss AI-driven health solutions. Recognizing oculars contributions, the PI was selected as an expert panelist for the AI for Social Good Award by the UAE government in collaboration with WHO, UNESCO, and Google. Ocular's work is backed by a \$1.5 million Google grant, reinforcing its mission to revolutionize healthcare diagnostics in Uganda. To ensure continued success, the project team also engaged in capacity-building training, equipping them with the skills needed to further AI integration in diagnosing Malaria, Tuberculosis, and Cervical Cancer.



The Ocular project team in a group photo with the Principal Prof. Tonny Oyana after the training on January 24, 2024, at Kampala's Golden Tulip Hotel.



Pics 1 and 2 of the oculars technical team showcasing during the Annual media convection at the Yusuf Lule auditorium ground



Science Week gets hands-on! Participants explore Ocular's innovative solution during the Science Technology and Innovation (STI) National Science Week



Ocular demonstrates disease diagnosis innovation at 27th Commonwealth conference



Showcasing our mobile microscopy solution at the Health Innovation Conference





The project team exhibited at the ITU GSR conference here in Uganda at the Speke Resort Munyonyo,

4.3 Visitors

Visitors play a vital role in showcasing the innovative work being done and fostering partnerships. Their presence allows for the exchange of ideas, offers opportunities for feedback, and helps build networks that can drive the project forward. Engaging with external visitors, including researchers, healthcare professionals, and potential collaborators, enhances the visibility of the lab's efforts and promotes broader support for AI-driven healthcare solutions. These interactions are essential for scaling the project's impact and encouraging cross-disciplinary collaboration. This section highlights some of the visitors hosted.

4.3.1 FPAIC Oversight Visit



FPAIC Committee members in a group photo with CoCIS staff after the meeting

On August 16, 2024, Makerere University Council's Finance, Planning, Administration, and Investment Committee (FPAIC) met with the management and administrative staff of the College to review the 2023/2024 financial year budget. Led by Chair Mr. Bruce Balaba Kabaasa, the meeting assessed budget utilization, compliance, and financial management improvements. Discussions focused on fund allocation, expenditure performance, and grant management. Kabaasa emphasized the need to verify that allocated funds facilitated tangible developments, such as equipment purchases and training programs, and stressed adherence to established grant management frameworks.

4.3.2 Director of Africa Initiative visits CoCIS Labs

Rachid Serraj, Director of Africa Initiative and Strategy for agriculture, water, and climate at Mohammed VI Polytechnic University (UM6P) in Morocco, visited different laboratories at CoCIS.



Mak AI Health Lab technical team showcased their AI-powered ocular tool on July 4, 2024.



Serraj, with a distinguished background in agriculture and research, previously served as the delivery manager for the FAO's Regional Initiative on Small Scale Family Farming and Rural Transformation in the Near East and North Africa. He has held several senior positions in international research organizations and is a prolific author with extensive experience in crop science, agronomy, and agri-food systems innovation.



Top and bottom the director visited the WIMEA and AirQo labs

4.3.3 Germans seek collaboration

A team of German scholars under the DAAD project coordinated by the institute of Languages and the German embassy visited the college to increase collaboration and research. The two met with researchers and discussed areas of interest.



4.3.4 Kakiri Senior Secondary students visit Mak AI Health Lab

The Makerere AI Health Lab hosted a group enthusiastic student from Uganda Martyrs SS Kakiri.



4.3.5 Nabinsunsa Girls Secondary School visit

The ocular team showcased the innovative mobile microscopy solution to the students of Nabinsunsa Girls who visited the university during their Career Day program. Oculars talented team of interns took the stage to demonstrate the ideation and 3D printing process behind the phone-to-microscope adapter.



5.0 | The Staff Environment

5.1 Human Resources

The college has three categories of staff namely: Academic, Administrative and Support. The College has several service departments that focus on: ICT Support Services; Innovations and Software Development; Finance; Registry; Research; Human Resource; International/Corporate/Public Relations; and Quality Assurance. The tables below provide a breakdown of these categories as of November 2024

The College of Computing and Information Sciences is Comprised of Academic, Administrative and Support staff whose numbers are indicated below.

SN	Category of staff	Filled Position
1	Academic staff	84
2	Administrative staff	18
3	Support staff	35
Total		137

5.1.2 Staff Promotions, Appointments and Confirmations in 2024

During the year 2024 the Human Resources Unit handled a number of staff – related matters regarding promotions, confirmations, retirements, study leave and transfers. Six (6) members of staff retired from work, three (3) were promoted to Lecturer position and One (1) promoted to Senior Lecturer position, six (6) members were transferred to Cocis, three (3) completed their study leave and four (4) were confirmed into the University service. This information is presented on the table 6 below.

Table 6 Staff Promotions, Appointments and Confirmations in 2024

No	Name	Position	Retirement Date
	Retired Staff		
1	Mr. Godfrey Kato	Ass. Accountant	Retired on 21/01/2024
2	Mr. William Ssebanakitta	Ass. Accountant	Retired on 20/02/2024
3	Mr. Patrick Feni Babanga	Ass. Accountant	Retired on 17/03/2024
4	Mr. Christopher Eryenga	Security Guard	Retired on 15/12/2024
5	Mr. Anthony Kwizera	Lab Attendant	Retired on 23/09/2024
6	Mr. Ronald Mwanje	CIPSD	Resigned Jan 2024
	Promotion		
7	Dr. Margaret Nagwovuma	Lecturer	Promoted from Assistant Lecturer to Lecturer on 1st July 2024
8	Prof. Engineer Bainomugisha	Professor	Promoted from Assoc. Prof. to Prof. on 25th Feb 2025
9	Dr. Emmanuel Mugejjera	Lecturer	Promoted from Assistant Lecturer to Lecturer on 1st July 2024
10	Dr. Daudi Jjingo	Senior Lecturer	Promoted from Lecturer to Senior Lecturer 1st July 2024
11	Dr. Dianah Kacunguzi Twinoburyo	Lecturer	Promoted from Assistant Lecturer to Lecturer 01/07/2024
12	Dr. Florence N Kivunike		Promoted from Lecturer to Senior Lecturer on 25th/2/2025
13	Dr. Odongo Stephen Eyobu	Senior Lecturer	Promoted from Lecturer to Senior Lecturer on 25th/2/2025
14	Dr. Drake Patrick Mirembe	Senior Lecturer	Promoted from Lecturer to Senior Lecturer on 25th/2/2025
	Transfers		
12	Ms. Ruth Iteu Eyoku	Principal Registrar	Transferred to CoCIS 21/06/2024
13	Ms. Madrine Namazzi Kayima	Administrative Assistant	Transferred to CoCIS 21/06/2024
14	Mr. Michael Hajusu	Ass. Accountant	on 21/01/2024
15	Mr. Arthur Asimwe Kampikaho	Custodian	Deployed to EASLIS on 16/05/2024
16	Ms. Sarah Nabawanuka	Ass. Accountant	Transferred to CoCIS 22/02/2024
17	Mr. Robert Ayesiga	Security Guard	Transferred to CoCIS on 28/ Sept/2024

Return From Study Leave			
18	Dr. Innocent Ndibatya	Lecturer	29/02/2024
19	Dr. Mark Abraham Magumba	Lecturer	31/07/2024
20	Mr. Paddy Junior Asimwe	Ass. Lecturer	30/11/2024
Confirmation			
21	Mr. Habert Baluku	Computer Lab Assistant	16/04/2024
22	Mr. Nicholas Betungye	Computer Lab Assistant	31/07/2024
23	Ms. Astaluzi Masika	Computer Lab Assistant	31/07/2024
24	Dr. Moses Golooba	Lecturer	31/07/2024

5.1.3 Volunteer Lecturer received from Nigerian Technical Aid Corps (TAC) program

This year Makerere University's College of Computing and Information Sciences (CoCIS) warmly welcomed Dr. Alo Oluwaseum Olubisi, a senior lecturer from Nigeria, who will serve as a volunteer in teaching, research, mentorship, and knowledge-sharing. Dr. Olubisi, assigned to the Department of Computer Science, was oriented into the college to familiarize herself with the environment. Her appointment is part of a broader effort led by Makerere University's Vice Chancellor to join the Nigerian Technical Aid Corps (TAC); a program facilitated by Nigeria to support partner African countries. His Excellency, the High Commissioner of Nigeria to Uganda, was instrumental in helping Makerere access this initiative, which aims to bolster academic expertise across African institutions.

Dr. Alo Oluwaseum Olubisi
meetings head Computer
Science Assoc. Prof.
Engineer Bainomugisha



5.1.4 Recognitions and Awards

5.1.4.1 Retired CoCIS Registrar and Accountants Honored

CoCIS has a culture of appreciating and celebrating staff who retire or leave the College honorably. This year we celebrated the services rendered by Retired

Principal Registrar Patrick Feni Babanga and two Assistant Accountants-Mr. Godfrey Kato and Mr. William Ssebanakkitta. The tradition of celebrating and appreciating retiring staff, is one, that has been cherished by the College leadership. Mr. Feni, Mr. Kato and Mr.Ssebanakkitta were recognized for their dedicated service to the university at a farewell luncheon that was presided over by the College Principal, Prof. Tonny Oyana at Piato Restaurant in Kampala. The three were presented with plaques and cash gifts. The event was also attended by unit heads and administrative staff.



College Principal, Prof. Oyana and the Deputy Principal CoCIS Dr. Nabende Cut Cake at the appreciation event for the retired staff

5.1.4.2 Ten staff get Vice Chancellor's Research Excellence Awards 2024

During this third session of the 75th Graduation Ceremony, the Vice Chancellor recognized the best researchers listed and published in the Graduation Booklet and the Mak News Magazine. The awards were based on Scopus database. The researchers were honored by the Vice Chancellor and Chairperson of Makerere University during the convocation luncheon held at Makerere University Convocation House. CoCIS best researchers included.

- Ass. Prof. Bainomugisha Engineer
- Ass. Prof. Nabukenya Josephine
- Dr. Nabende Peter
- Dr. Odongo Eyobu Steven
- Mr. Marvin Ggaliwango
- Dr. Mwebaze Johnson
- Dr. Namatovu Hasifah Kasujja
- Dr. Jjinga Daudi
- Dr. Nakatumba Joyce Nabende
- Dr. Namujjuzi Sylvia

During the graduation Ceremony, College of Computing and Information Sciences was further Commended for Noble Research in Artificial Intelligence to Solve Africa's Problems

5.1.4.3 Dr. Rose Nakasi Awarded during Trend Setter Business forum



Nakasi won the prestigious Business Trends Pace Setter award during the 5th Business Trendsetters Forum 2024 held on 29th August 2024 at Sheraton Hotel in Kampala organized by The Business Trends Galore and partners moderated by Ms. Yvonne Mpanga, a Business Development Consultant. The event centered around the theme "Artificial Intelligence, Your Mind, and Your Business." The

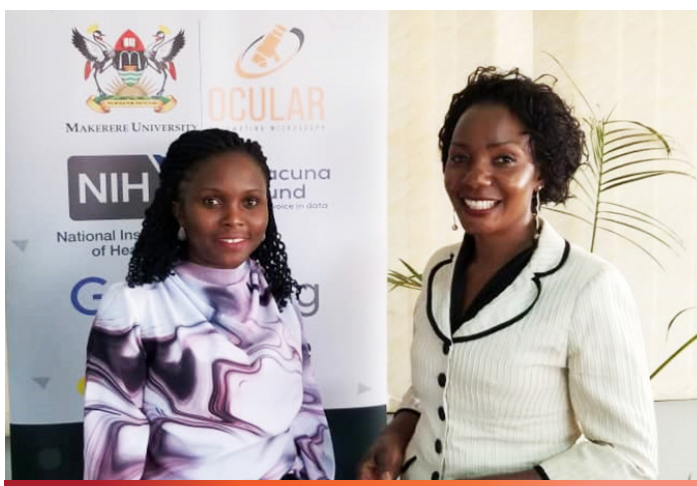
event brought together notable figures, who discussed the topical issues of the day to forge a way forward for business as they utilize new technologies in the medical, health care and others industries.



Rose Nakasi Team Leader, Ocular AI Project, Makerere University.

Rose Nakasi is a distinguished Artificial Intelligence (AI) researcher whose work is at the forefront of technological innovation in healthcare. Her research interests lie at the intersection of Computer Vision and Machine Learning, specifically focusing on applying these cutting-edge technologies for automated microscopic diagnosis and spatial modeling of diseases in low-resourced countries. Her research interest is artificial intelligence, particularly in developing low-cost tools for improved automated solutions for diagnostic health challenges, such as tools to support microscopy diagnostics for diseases like malaria, cervical cancer, tuberculosis, and intestinal parasites.

**Dr. Rose Nakasi and
Yvonne Mpanga at
Makerere AI- Health Lab**



5.1.4.4 Staff establishment structure as of December 2024

Department	Established Positions	Filled Positions	Male	Female	Vacant Positions
Computer Science	21	12	8	4	9
Information Systems	21	16	5	11	11
Information Technology	14	19	8	11	0
Networks	21	17	12	5	5
Records and Archives	14	7	4	3	7
Library and Information Science	21	13	4	9	7
Total	112	84	41	43	39

School of Computing and Informatics Technology Established and filled positions for academic staff in the Department of Computer Science (CS)

Position	Established Positions	Filled Positions	Male	Female	Vacant Positions
Professor	5	1	1	0	4
Associate Professor	5	0	0	0	5
Senior Lecturer	5	3	1	1	2
Lecturer	4	3	1	2	1
Assistant Lecturer	2	5	4	1	0
Teaching Assistant	2	0	0	0	2
Research Fellow	2	0	0	0	2
Stenographer secretary	1	0	0	0	1
Office Attendant	1	0	0	0	1
Total	27	13	7	4	14

Established and filled positions for academic staff Department of Information Systems (IS)

Position	Established Positions	Filled Positions	Male	Female	Vacant Positions
Professor	5	0	0	0	5
Associate Professor	5	3	0	3	2

Senior Lecturer	5	3	1	2	2
Lecturer	4	6	2	4	0
Assistant Lecturer	2	4	2	2	0
Teaching Assistant	2	0	0	0	2
Research Fellow	2	0	0	0	2
Stenographer secretary	1	0	0	0	1
Office Attendant	1	0	0	0	1
Total	27	16	5	11	15

Established and filled positions for academic staff Department of Information Technology (IT)

Position	Established Positions	Filled Positions	Male	Female	Vacant Positions
Professor	3	0	0	0	3
Associate Professor	3	0	0	0	3
Senior Lecturer	3	3	0	3	0
Lecturer	2	6	5	7	0
Assistant Lecturer	2	5	3	2	0
Teaching Assistant	2	0	0	0	2
Research Fellow	2	0	0	0	2
Stenographer secretary	1	0	0	0	1
Office Attendant	1	0	0	0	1
Total	20	14	8	12	12

Established and filled positions of academic staff in the Department of Networks

Position	Established Positions	Filled Positions	Male	Female	Vacant Positions
Professor	5	0	0	0	5
Associate Professor	5	1	1	0	4
Senior Lecturer	5	5	4	1	0
Lecturer	4	10	7	3	0
Assistant Lecturer	2	1	1	0	0
Teaching Assistant	2	0	0	0	2
Research Fellow	2	0	0	0	2
Stenographer secretary	1	0	0	0	1
Office Attendant	1	0	0	0	1
Total	27	17	13	4	15

Academic staff establishment in the East African School of Library and Information Sciences

Established and filled positions for academic staff in the Department of Records and Archives

Position	Established Positions	Filled Positions	Male	Female	Vacant Positions
Professor	3	1	1	0	2
Associate Professor	3	1	1	0	2
Senior Lecturer	3	0	0	0	3
Lecturer	3	2	0	2	1
Assistant Lecturer	4	3	2	1	1
Teaching Assistant	2	0	0	0	2
Research Fellow	2	0	0	0	2
Stenographer secretary	1	0	0	0	1
Office Attendant	1	0	0	0	1
Total	22	7	4	3	15

Established and filled positions for academic staff in the Department of Library and Information Science

Position	Established Positions	Filled Positions	Male	Female	Vacant Positions
Professor	5	1	1	0	5
Associate Professor	5	0	0	0	4
Senior Lecturer	5	2	0	3	2
Lecturer	4	1	0	0	4
Assistant Lecturer	2	9	2	7	0
Teaching Assistant	2	0	0	6	2
Research Fellow	2	0	0	0	2
Stenographer secretary	1	0	0	0	1
Office Attendant	1	0	0	0	1
Total	27	13	3	16	21

5.1.4.5 Details of Academic Staff Qualifications and research interests

Department of Computer Sciences

SN	Name	Title	Highest Qualification	Awarding institution (s)	Area of specialization
1	Engineer Bainomugisha	HoD, Professor	PhD	Vrije Universiteit Brussels	Programming Language Engineering Cloud & Mobile Computing Secure Programming Distributed Systems Artificial Intelligence
2	John Ngubiri	Senior Lecturer	PhD	University of Groningen	Software Security Parallel Systems Process Modelling Algorithms
3	Joyce Nabende Nakatumba	Senior Lecturer	PhD	Eindhoven University of Technology	Data & Process Mining Business Process Management Process Modelling Artificial Intelligence and Machine learning
4	Daudi Jjingo	Lecturer	PhD	Georgia Institute of Technology	Bioinformatics of Infectious and Chronic Diseases Health Informatics Artificial Intelligence
5	Michael Kizito	Lecturer	PhD	University of Gothenburg	IT Governance Computer Security Digitalization Health Informatics
6	Marriette Atuhuriire Katarahweire	Lecturer	PhD	Makerere University	Computer Security Privacy Software Engineering Mobile health
7	Rose Nakibuule	Lecturer	PhD	Makerere University	Computer Vision Image processing Artificial Intelligence Data Science
8	John Kizito	Assistant Lecturer	MSc	University of Groningen	Distributed Systems Programming Languages Theoretical computing

9	Swaib Dragule	Assistant Lecturer	MSc	University of Mysore	Robotics Data structures and algorithm's Programming languages
10	Barbara Nansamba	Assistant Lecturer	MSc	Makerere University	Computer Architecture Computer Networks and Data communications Theoretical computing
11	Emmanuel Lule	Assistant Lecturer	MSc	Makerere University	Computer Networks Operating systems Internet of Things
12	Marvin Ggaliwango	Assistant Lecturer	MSc	BRAC University	Responsible Artificial Intelligence systems Health systems

Department of Information systems

SN	Name	Title	Highest Qualification	Awarding institution (s)	Area of specialization
1	Agnes Rwashana Semwanga	Assoc. Prof.	PhD	Makerere University	Information Systems, Systems Modeling, Health Informatics
2	Josephine Nabukenya	Assoc. Prof.	PhD	Radboud University Nijmegen, Netherlands	Health Informatics, Information Systems, Enterprise Architecture, Business Process Management
3	Rehema Baguma	Assoc. Prof.	PhD	Radboud University Nijmegen, Netherlands	Human Computer interaction, digital inclusion, responsible/ethical AI & data governance
4	Agnes Nakakawa	HoD, Senior Lecturer	PhD Computer Science	Radboud University Nijmegen, Netherlands	Human Computer interaction, digital inclusion, responsible/ethical AI & data governance
5	Emily Bagarukayo	Senior Lecturer	PhD	Radboud University Nijmegen, Netherlands	Higher order cognitive skills development, eLearning, eservices, ICT4D, security
6	Peter Nabende	Senior Lecturer D/Principal	MSc	University of Groningen, Netherlands	Computational Linguistics/ NLP; Intelligent Systems/ AI; Data Mining

7	Peter Wakholi	Lecturer	MSc	University of Bergen, Norway	Business Process Management, Health Information Systems, Enterprise Systems, System Design
8	Namatovu Hasifah Kasujja	Lecturer	MSc	University of Groningen, Netherlands	Spatial analysis, Health Informatics, Water Informatics, Systems Dynamics, Information Systems Analysis and Design.
9	Grace Kobusinge	Lecturer	Ph.D in Informatics	University of Groningen, Netherlands	Health Informatics, Information Systems Interoperability, Human computer Interaction, ICT4D
10	Arinaitwe Irene	Lecturer	Ph.D	Makerere University	GIS, Persuasive technology, E-government and e-participation, system modelling
11	Arinaitwe Irene	Lecturer	Ph.D	Makerere University	GIS, Persuasive technology, E-government and e-participation, system modelling
12	Mercy Rebekah Amiyo	Lecturer	Ph.D	University of Groningen, Netherlands	Information Systems Design, Business Process Management
13	Mark Abraham Magumba	Lecturer	Ph.D	Makerere University	Machine Learning / Artificial Intelligence/ Natural Language Processing / Health Informatics
14	Katende Jacob	Assistant Lecturer	MSc. IS	University of Groningen, Netherlands	Information Systems, Computer Networks, AI/ NLP/Deep Learning
15	Asiimwe Paddy Junior	Assistant Lecturer	MSc	Makerere University	Information Systems, Computer Networks, Localization, IoT, Wireless Sensor Networks, Mobile Application Development, Embedded systems

16	Fiona Ssozi	Assistant Lecturer	MSc. IS	London South Bank University	Information Systems Design, Human computer Interaction, Participatory Design
17	Asio Evelyn Patra K	Assistant Lecturer	MSc	Korea Advanced Institute of Science and Technology	Knowledge Service Engineering: Data Science, Knowledge systems, Information Systems, Human computer Interaction, Intelligent User Interface design, Interaction Design, User centered Design, Health Informatics, Human factors, ergonomics and Cognitive Engineering,

Department of Information Technology (IT)

SN	Name	Title	Highest Qualification	Awarding institution (s)	Area of specialization
1	Tulinayo P. Fiona	HoD, Senior Lecturer	PhD	Radboud University	Method engineering, Information Systems Engineering, Causal reasoning, System dynamics, domain and conceptual modelling
2	Evelyn Kigozi Kahiigi	Senior Lecturer	PhD	DSV, Stockholm University	E-learning, ICT4D and Digital Transformation
3	Florence Nameere Kivunike	Senior Lecturer	PhD	DSV, Stockholm University	Development Informatics - Digital Government, Health; Ethics in informatics
4	Annabella H. D. Basaza	Lecturer	PhD	University of Groningen (PhD)	Digital Transformation, Health Informatics, Machine Learning, Evaluation studies, Mining SME's, Information Security, Decision Enhancement Services, AI

5	Golooba Moses	Lecturer	MSc	International Islamic University, Malaysia	AI, IoT, and their applications within: e-Health, e-Agricultural, e-Learning, e-Governance and Cyber Security.
6	Namirembe Esther	Lecturer	MSc	University of Cape Town	E-learning and ICT for development
7	Hawa Nyende	Lecturer	PhD	University of Gothenburg	Informatics- Health, Service Innovation , web systems, data warehousing and business intelligence, Ethics in IT
8	Innocent Ndiabata	Lecturer	PhD	Stellenbosch University	Sustainable and Intelligent Transport Systems, Agent-Based Modeling, Artificial Intelligence
9	Chongomweru Halimu	Lecturer	PhD	Universiti Teknologi Brunei (UTB)	Artificial Intelligence, Machine Learning and Data Science
10	Alice Nandawula Mugisha	Lecturer	PhD	University of Bergen	Digital health systems, systems modelling and Health Data Science
11	Raymond Mugwanya	Lecturer	PhD	University of Cape Town	Digital Health, ICT4D, Human Centered Design, Digital Transformation
12	Nagwovuma Margaret	Lecturer	PhD	Makerere university	eServices, performance evaluation, ICT4D, digital Innovation and ethics
13	Bitwire George Albert	Assistant Lecturer	Masters	Makerere university	Artificial Intelligence, Data Science, Web system development
14	Komugisha Lilian	Assistant Lecturer	Masters	Makerere university	Collaboration Engineering
15	Emmanuel Mugejjeru	Lecturer	Masters	Makerere university	ICT in Agriculture, Computer Programming, Business Intelligence.
16	Male Vincent	Assistant Lecturer	Masters	Nkumba University	Web based systems and services, Data Analytics, Network security
17	Flavia Namagembe	Assistant Lecturer	Masters	Makerere university	Digital health, Implementation Science, Enterprise Architecture

18	Rose Nakasi	Lecturer	PhD	Makerere university	Computer Science
19	Serugunda Henry Mukalazi	Assistant Lecturer	MSc , MBA	Makerere university	Health Systems and Artificial Intelligence

Department of Networks

SN	Name	Title	Highest Qualification	Awarding institution (s)	Area of specialization
1	Benjamin Kanagwa	Associate Professor	PhD	Radboud University Nijmegen, Netherlands	Software engineering, Software architectures, Cloud computing, AI
2	Swaib Kyanda Kaawaase	Senior Lecturer	HoD, Senior Lecturer	Harbin Engineering University, China	Wireless networks, wireless network protocols, wireless network security
3	Julianne Sansa Otim	Senior Lecturer	PhD	Radboud University Nijmegen, Netherlands	Internet of things for developing regions, ICT4Development Communications network protocol design, Internet and Telecommunication policies, Quality of service and Quality of experience, Systems security
4	Joseph Kikombo Balikuddembe	Dean, Senior Lecturer	PhD	University of Cape Town	Software Development, Systems Analysis and Design, Software Sizing & Estimation, Requirement's engineering, Strategic Project Management, Business Intelligence
8	Moses Ntanda Kyebambe	Lecturer	PhD	Xiangtan University, China	Machine learning, deep learning, data science
9	Tonny Eddie Bulega	Lecturer	PhD	South China University of Technology, Guangzhou China	Mobile wireless networks, wireless sensor networks, communication systems

10	Ruth Mbabazi Mutebi	Lecturer	PhD	Makerere University	Persuasive technologies, wireless networking, Requirements engineering
11	Johnson Mwebaze	Lecturer	PhD	Radboud University Nijmegen, Netherlands	Data communications and networks, scientific data processing, web and computer programming
12	Drake Patrick Mirembe	Lecturer	PhD	University of Groningen, Netherlands	Mobile and wireless technologies, Cyber security, Requirement's engineering, Software systems design and integration, educational technologies, ICT innovation and Incubation, Policy and Strategy, ICT4Development, Web Technologies, Data science
1	Joab Ezra Agaba	Lecturer	PhD	Makerere University	Software Development, re-usability in instructional design for e-learning environments
1	Odongo Steven Eyobu	Lecturer	PhD	Kyungpook National University, South Korea	Deep learning Systems, Localization and Navigation, Vehicular Communications, Internet of Things, Intelligent Transportation Systems
1	Nasser Kimbugwe	Lecturer	PhD	Xiangtan University, China	AI (Machine learning and Deep learning), software engineering, internet of things, programming languages, data science
1	Mary Nsabagwa	Lecturer	PhD	Makerere University	Wireless sensor networks and embedded systems
1	Rashidah Kasauli Namisanvu	Lecturer	PhD	Chalmers University of Technology, Sweden	Software engineering, requirements engineering, large-scale systems development, safety-critical systems development, big data analytics, communications engineering, agile software development

1	Benard Muwonge Sajjabbi	Lecturer	PhD	Xiangtan University, China	Optimization of communication Networks, Cognitive Radio Networks, Multimedia communication systems, Computer Networks and system security, Tele-health and telemedicine, IoT
1	Grace Bugembe Kamulegeya	Lecturer	PhD	Makerere University	Software architectures, Business Process Management, Software Innovation, Software Innovation commercialization, software start-ups
1	Alex Mwotil	Assistant Lecturer	Master's	Makerere University	Computer networks, cloud computing, distributed systems

Department of Records and Archive Management

SN	Name	Title	Highest Qualification	Awarding institution (s)	Area of specialization
1	Magara Elisam	Professor	PhD	University of South Africa UNISA	Records and Archives Management
2	Luyombya David	HoD, Assoc. Professor	PhD	University College London, UK	Records and Archives Management
3	Namujuzi Sylvia	HoD, Assoc. Professor	PhD	Makerere University	Records and Archives Management
4	Kacunguzi Twinoburyo	Lecturer	PhD	University of Illinois, USA	Records and Archives Management
5	Kiconco Christine	Assistant Lecturer	Masters	Uganda Martyrs University	Records and Archives Management
6	Ekwaro Francis	Assistant Lecturer	Masters	Makerere University	Records and Archives Management

Department of Library and Information Science

SN	Name	Title	Highest Qualification	Awarding institution (s)	Area of specialization
1	Okello-Obura Constant	Professor	PhD in Information Science	University of South Africa, Pretoria	Business Information Systems and Services; Information Science Education; Records and Information System Security; Bibliometrics; Indigenous Knowledge Management Systems; User Studies
2	Bukirwa Joyce Muwanguzi	Senior Lecturer	PhD in Information Science	Makerere University	Automation of Library and Information Systems, Digital Libraries, Information Processing, Management of Electronic Records/ Documents, Grey Literature Studies
3	Kaddu Sarah Birungi	Dean, Senior Lecturer	PhD in Information Science	Makerere University	Information Storage & Retrieval, Information Ethics, Management of information institutions
4	Agabirwe Patience	Assistant Lecturer	Master's of Library and Information Science-Digital library learning	Oslo Metropolitan University, formerly Oslo University college and Arkeshus University Norway	Knowledge organization and management (Cataloguing, classification and Web content Management), information technology applications in libraries, information storage and retrieval, Digital librarianship, Automation of libraries, special libraries, National, public and community libraries, information practices, information seeking behavior, information security and management, universal access to information.

6	Amanda Irene Birungi	Assistant Lecturer	Master's Degree in information services	London Metropolitan university , UK	Library automation, library services for children and young people, Library science
7	Batte Richard	Assistant Lecturer	Master's Degree in information services	Makerere University	Publishing, Systems Dynamics and ICTs for development
8	Kutyamukama Alice Gitta	Assistant Lecturer	Master's Degree in information services	Makerere University	Health information access, information service, information literacy, digital libraries, ICTs in libraries, information entrepreneurship, applied organization psychology
9	Kulisooma Ezera	Assistant Lecturer	Master's of Library and Information Science-Digital library learning	Oslo Metropolitan University, formerly Oslo University college and Arkeshus University Norway	Information and Knowledge Management, Information and Digital literacy, Digital librarianship
10	Mutibwa Lois Nankya	Assistant Lecturer	Master's in Library and information Science	Makerere University	Museum and heritage studies; information resources and services, Social media, Libraries & Librarianship. Children literature & services. Children literature & services.
11	Muzaki Faridah	Assistant Lecturer	Master's of science in information science, Master of science in computer science	Makerere University	Information and Digital Literacy, Internet of Things, ICTs in library service
12	Mwanje Ssenono Aloysius	Assistant Lecturer	MSC in information	Mahati Magandhi university,	Data management, Data literacy, Wed programming, knowledge

			and computer	Kottayan Kerala, India	organization, Knowledge and Information Management, Information Architecture, Information Systems Analysis and Design
13	Nalumansi Mary	Assistant Lecturer	Master's of science in information science	Makerere University	Information & Society, Information Literacy, Communication Skills & Practice and Reading Culture

5.1.4.6 Administrative and Support staff

No	Administrative staff units	Title
	Principal's Office	
1	Prof. Tonny Oyana	Principal
2	Julius Lebo	Principal Human Resources Officer
3	Angida Franco Mugyema	Accountant
4	Micheal Hajusa	Assistant Accountant
5	Jane Anyango	Principal Communications Officer
6	James Isagara Kisoro	Web Administrator
7	David Ikoma	Procurement Officer
8	Clare Nata Musoke Bukenya	Senior Administrative Assistant
9	Dora Bampangana	Research Grants Officer
10	Viola Kemanzi	Administrative Assistant
11	Viola Kemanzi	Administrative Secretary
12	Pearl Arinda	Administrative Assistant
	Registrar's Office	
13	Ruth Iteu Eyoku	Principal Registrar
14	Grace Apili	Assistant Registrar
15	Kabuye Mathias	Administrative Assistant
16	Ayebare Maureen Liliane	Administrative Assistant
17	Kintu Samuel	Clerical Officer
	Support Staff Units	
18	Hope Nahwera	Chief-Custodian
19	Richard Kananura	Custodian
20	Arthur Asimwe Kampikaho	Custodian
21	Victor Mujungu	Chief-Custodian

22	Federesi Namutebi	Office Attendant
23	Prossy Ainomugisha	Office Attendant
24	Gorret Nantale	Office Attendant
25	Immaculate Sanyu	Office Attendant
26	Rehema Nalwadda	Office Attendant
27	Peter Byarugaba	Office Attendant
28	Jolly Mukasa	Office Attendant
29	Sulaiman Munyogoli	Security Guard
30	Godfrey Adima	Security Guard
31	Christopher Eryenga	Security Guard
32	William A. Kirunda	Lab. Attendant
33	Rebecca Kalyowa	Lab. Attendant
34	Christine Ochokoru	Lab. Attendant
35	Zziwa Sarah	Lab. Attendant
36	Sarah Amanyire	Lab. Attendant
37	Betty Kyeyune	Lab. Attendant
38	Christine Akankwasa	Lab. Attendant
39	Irene Tasumba	Lab. Attendant
40	Florence Nabatanzi	Lab. Attendant
41	Joseph Afayo	Office Attendant
42	Editta Ayebare	Office Attendant
43	Habert Baluku	Senior Assistant Technician
44	Nicholas Betungye	Senior Assistant Technician
45	Claire Wessaali	Computer Lab.Assistant
46	Masika Astaluzi	Senior Assistant Technician
47	Edward Zziwa	Technician
48	Richard Kayiggwa	Office Attendant
49	Badiru Senkumba	Assistant systems Administrator
50	Phiona Costance Timugibwa	Assistant systems Administrator
51	Christine Nanyonga	Records Assistant

6.0 | Recent Development

6.1 Launch of the Mak AI -Health Lab

The college has three categories of staff namely: Academic, Administrative and Support. The College has several service departments that focus on: ICT Support Services; Innovations and Software Development; Finance; Registry; Research; Human Resource; International/Corporate/Public Relations; and Quality Assurance. The tables below provide a breakdown of these categories as of November 2024

The College of Computing and Information Sciences is Comprised of Academic, Administrative and Support staff whose numbers are indicated below.



Prof. Edward Bbaale, Dr. Rose Nakasi, PS Dr. Amina Zawedde and Prof. Tonny Oyana after unveiling the lab.

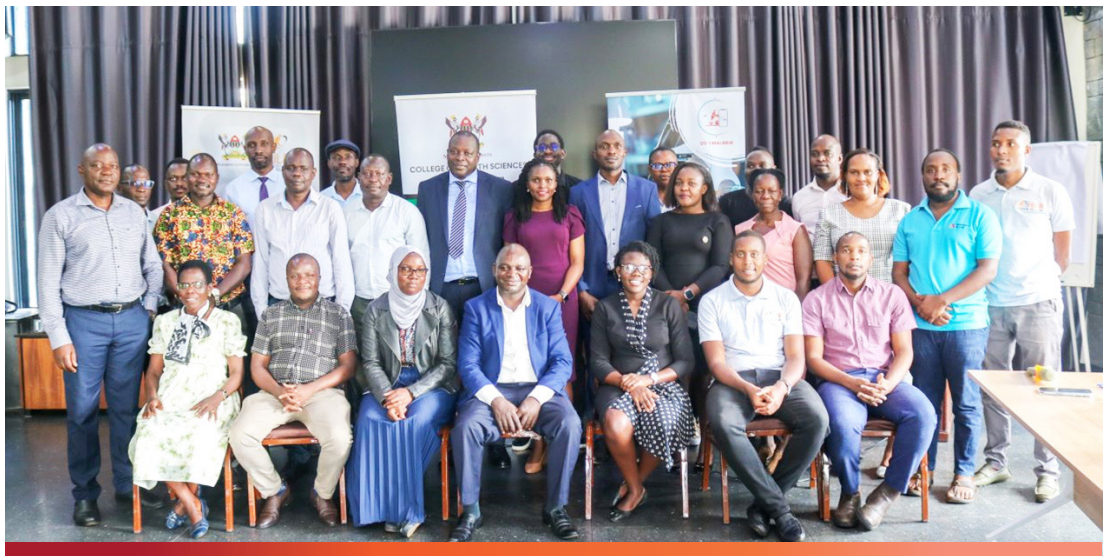
In a landmark collaboration, the Government of Uganda and Makerere University inaugurated the Artificial Intelligence Health Lab, located in Block B of the College of Computing and Information Sciences. This pioneering initiative aims to revolutionize healthcare delivery in Uganda through the transformative capabilities of artificial intelligence (AI), marking a significant stride towards technological advancement in the nation's healthcare sector.

Launching the lab on May 30, 2024, Dr. Amina Zawedde, Permanent Secretary of the Ministry of ICT and National Guidance, underscored the importance of this development. "Today marks a momentous occasion as we unveil the Artificial Intelligence Health Lab at Makerere University.

This milestone signifies not just a step forward for the university but for our entire nation as we enter an era of innovation and advancement in healthcare through artificial intelligence," she remarked.

Dr. Zawedde emphasized AI's potential to revolutionize healthcare, including diagnostics, treatment planning, and personalized care. She highlighted ongoing government initiatives to integrate AI into healthcare, such as telemedicine platforms, health data analytics, and AI-driven solutions for medical imaging and drug discovery. This initiative reflects Uganda's commitment to leveraging cutting-edge technology to enhance healthcare outcomes and meet the evolving needs of its population.

6.2 DS-I Malaria Project Launched



The DS-I Malaria Project, aimed at revolutionizing malaria diagnosis and surveillance in Uganda, was officially launched at Fairway Hotel in Kampala on March 11, 2024. Funded by NIH under the Data Science Africa Initiative, the project is a collaborative effort involving Makerere University's Mak AI Health Lab, the College of Health Sciences, Kiruddu National Referral Hospital, Boston University, Mulago Hospital, Sunbird AI, the Ministry of Health, and the Okina Project. The initiative seeks to leverage Artificial Intelligence (AI) for automated microscopy, addressing the critical shortage of trained lab technicians and enhancing disease detection capabilities. Led by Dr. Rose Nakasi and launched by Prof. Tonny Oyana of the College of Computing and Information Sciences, the DS-I Malaria Project focuses on developing a rapid, low-cost, and accurate in-field screening system

using AI-enhanced image analysis models. Dr. Nakasi emphasized the project's innovation, which includes a novel 3D printed adapter enabling smartphones to be integrated with microscopes, facilitating real-time diagnostics and surveillance of malaria across Uganda's diverse healthcare settings.

The launch highlighted field surveys conducted across Eastern, Central, Northern, and West Nile regions to assess infrastructure readiness and technician acceptance of the AI-driven microscopy application. With Malaria remaining a significant health and economic burden in sub-Saharan Africa, the DS-I Malaria Project aims to pioneer practical, field-deployable solutions that leverage cutting-edge AI technologies to improve disease management and public health outcomes in the region.

6.2.1 Center for Innovations and Professional Skills Development (CiPSD)

Ms. Nalubega Barbra.
Head CIPDS



The Center for Innovations and Professional Skills Development (CiPSD) started off as a Cisco Networking Academy under the then faculty of computing and Information Technology in 2004. It then expanded into a highly regarded unit in Makerere University that plays a significant role in providing individuals with practical computer knowledge and skills to support a 21st century knowledge-based economy. The center strategically offers training that is responsive to the needs of society through tailoring courses towards what the industry demands. The Center has been at the forefront in assisting government and other organizations to build human capacity with various ICT skills.

In a quest to respond to industry demands, CiPSD runs the following short professional courses: The Microsoft office Suite (CCA), CCNA, CCNP, Oracle 19c, IC3 Digital Literacy, Cyber Security Operations, Ethical Hacking, Programming in Python, 3D Animation, DEVNET, Kubernetes, 5G Technology, Dynamic Web development, Data Analysis& Visualization using Python, Microsoft AZURE CLOUD and Advanced PC repair and soft maintenance, Artificial Intelligence and many more. And in the last year over 1,223 learners have been impacted by the courses on offer as well as through boot camp initiatives.

Furthermore, during the University's recess term, CiPSD takes charge of

the hands-on sessions in the CCNA course, the Systems administration course focusing on Linux in support of academic programs for students pursuing the BSc Computer Science and BIST courses.

To augment the Professionalism, CiPSD as an authorized Global Testing Center on behalf of CoCIS hosts the Pearson VUE, Prometric, GMAT, IC3, ICDL, ISACA/PSI and Kryterion Certification Examinations. A significant number of Professionals from all over East and Central Africa sit their international certification exams at the Center because of the strong, credible and successful track record of more than twenty years. (This year has registered an increase in the number of clients testing with us because we offer excellent and credible service).

CiPSD Manages Consultancy Projects and other Government programs like the Evbab project that runs Undergraduate and Graduate Programs for 1,680 students on behalf of the Government of Uganda and that of India, which programe is running smoothly and will officially close in 2027.(We anticipate to have at least 500 learners who will be graduating in August 2025 from this project)

As CiPSD consultants, we also ran and successfully completed a UCC/ UCUSAF funded project titled 'Digital Skills Development of Rural Unserved & Underserved

Communities' comprising of 15 districts namely; Moyo, Kotido, Karenga, Otuke, Amudat, Nwoya, Lamwo, Agago, Amuria, Ngora, Pader, Ntoroko, Kagadi, Bundibugyo and Kanungu which project impacted the lives of 2,643 participants at the time of completion.

Under the Huawei ICT Academy Programs in partnership with Huawei Technologies Uganda Company Ltd., this year we have trained a total of 1,160 with participants taking on courses such as HCIA and 5G technology, I and Ethical Hacking as well.

CoCIS was approved as a Regional Cisco Networking Academy under a partnership with Cisco Systems Inc. to oversee the Cisco programs in the region and ensure that Local Academies operate in line with the CISCO Quality Assurance plan. Its target audience includes Cisco academies, students, public universities, primary and secondary schools and business entities. The main partner is Cisco Systems Inc.

Boot Camp Initiatives

Boot Camp initiatives at CoCIS provide intensive skill focused

training in high demand areas of technology. Through these initiatives we were able to provide students and professionals with the skills they need to succeed in the rapidly evolving technology. In 2024, we had a total of 658 participating in different trainings as indicated below.

- I. The Nov-Dec 2024 boot camp initiative attracted 329 attendees who took attended one of the following courses: Python full stack developer, Web design, Microsoft suite, Mobile apps developer, CCNA Networking, 3D computer animation, Videos editing and content creation.
- II. In June 2024, we had 111 participants who were trained in Cyber Security and Ethical hacking course. It was an intensive hands-on training for 3 hours per group from 4th June to Sunday 9th June 2024
- III. 70 participants turned up for the Azure cloud training in June 2024.
- IV. With respect to the Oracle 19c training 47 participants attended the training in the last week of July 2024.
- V. On the AI and Data Science programme 101 participated in the Aug /Sept intake of 2024



Training and Career Guidance for Secondary Schools.

This year we received a number of secondary schools with students who wished to tour our facilities and get career inspiration from there. We hosted Kampala Quality Primary School, Galaxy International School, Seeta High School, Mbogo Mixed Secondary School, Merryhill Christian School, Nabisunsa Girls School, and Daystar University (Kenya)

The Smart Classroom has offered a conducive environment for both teaching and learning, through conference tool integration and live broadcasting, the computers have up to-date software applications,

reliable internet connectivity (wired and wireless), controlled room temperature.

Users of the Smart Classroom have been equipped with various skill sets to enable them navigate the world with the ever-emerging ICT trends. Components like the robot can be programmed to perform automated tasks in both teaching and learning. The academic staff acquired knowledge on the different components of the smart classroom and how they assist in carrying out ICT- leveraged teaching and learning.



7.0 | Financial Report

Budget and Budget Performance - 6 Months

	Funds Available (UGX): Account	Established Positions		Encumbrance	Female	Vacant Positions
1	01-002-120100-30101 00300-0000-320036-0 000-02-221001	Advertisements	23,250,000	-	23,250,000	-
2	01-002-120100-30101 00300-0000-320036-0 000-02-221002	Workshops meetings and Seminars	14,000,000	-	13,539,888	460,112
3	01-002-120100-30101 00300-0000-320036-0 000-02-221017	Membership and subscription	1,625,000	-	-	1,625,000
4	01-002-120100-30101 00300-0000-320036-0 000-02-224011	Research Expenses	168,450,000	-	131,281,286	37,168,714
5	01-002-120100-30101 00300-0000-320043-0 000-02-211106	Allowances	138,192,034	-	137,699,134	492,900
7	01-002-120100-30101 00300-0000-320043-0 000-02-221001	Advertising and Public Relations	29,500,000	-	29,499,972	28
8	01-002-120100-30101 00300-0000-320043-0 000-02-221003	Staff Training	26,250,000	-	1,432,500	26,250,000
9	01-002-120100-30101 00300-0000-320043-0 000-02-221007	Books, Periodicals and Newspapers	5,700,000	-	1,432,500	4,267,500
10	01-002-120100-30101 00300-0000-320043-0 000-02-221008	Information and Computer Technology Supplies	66,350,000	-	66,349,008	992
11	01-002-120100-30101 00300-0000-320043-0 000-02-221009	Welfare and Entertainment	98,123,036	1,906,880	96,215,550	606
12	01-002-120100-30101 00300-0000-320043-0 000-02-221011	Printing, Stationery, Photocopying	25,000,000	-	24,988,320	11,680

13	01-002-120100-30101 00300-0000-320043-0 000-02-221017	Membership and subscrip tion	4,000,000	-	750,000	3,250,000
14	01-002-120100-30101 00300-0000-320043-0 000-02-222001	Information and Computer Technologies	15,000,000	-	7,500,000	7,500,000
15	01-002-120100-30101 00300-0000-320043-0 000-02-222002	Postage and Courier	1,500,000	-	1,127,492	372,508
16	01-002-120100-30101 00300-0000-320043-0 000-02-223001	Property Manageme nt Expenses	29,500,000	14,740,000	14,750,000	10,000
17	01-002-120100-30101 00300-0000-320043-0 000-02-223004	Guard and security services	18,500,000	-	17,584,000	916,000
18	01-002-120100-30101 00300-0000-320043-0 000-02-224008	Education and material services	296,327,000	11,664,300	248,058,668	36,604,032
19	01-002-120100-30101 00300-0000-320043-0 000-02-226001	Insurances	2,500,000	-	-	2,500,000
20	01-002-120100-30101 00300-0000-320043-0 000-02-227004	Fuel, Lubricants and oils	23,400,000	-	23,400,000	-
21	01-002-120100-30101 00300-0000-320043-0 000-02-228001	Maintenan ce - Building and structures	20,000,000	-	18,530,000	1,470,000
22	01-002-120100-30101 00300-0000-320043-0 000-02-228002	Maintenance - Transport Equipment	15,000,000	-	-	15,000,000
23	01-002-120100-30101 00300-0000-320043-0 000-02-228003	Maintenance - Machinery & Equipment	23,510,500	-	10,873,000	12,637,500
24	01-002-120100-30101 00300-0000-320043-0 000-02-228004	Maintenance - Other Fixed Assets	42,600,000	16,600,000	6,808,920	19,191,080
25	01-002-120100-30101 00300-0000-320043-0 000-02-282103	Scholarships and related costs	7,000,000	-	-	7,000,000
	TOTAL		1,095,277,570	44,911,180	873,637,738	176,728,652

8.0 | CoCIS in the Media

Media coverage is essential for raising public awareness, attracting potential collaborators, and enhancing the project's visibility. By showcasing college breakthroughs and impact, media platforms help inform the broader community about the benefits of AI in healthcare diagnostics. This coverage also plays a crucial role in engaging policymakers, healthcare professionals, and funders, driving support for the project's goals. Additionally, media attention amplifies the project's successes, fostering a positive reputation and inspiring confidence in its capacity to address pressing health challenges. In the period under review Ocular presence in social and conventional media was vivid.

Launch of the Ocular Website

Ocular's official website went live, serving as a central hub for information about the project's progress, media coverage, and partnership opportunities, further enhancing its credibility and outreach. Access the website at: <https://www.ocular-project.com/>

Makerere University Innovations Magazine 2024

Ocular AI Innovation was highlighted as one of the cutting-edge technologies

in the university. The application of AI in healthcare was described as one era where the university is making significant strides. The magazine was published by the New vision newspaper during the re-opening of the iconic building.

China Global TV interview about the Ocular project.

During her interview on China Global TV, Dr. Rose Nakasi discussed several critical aspects of her AI-driven health diagnostics project at Makerere University. She provided an overview of the Ocular project, emphasizing its mission to leverage AI technology for diagnosing Malaria, Tuberculosis, and Cervical Cancer. Dr. Nakasi highlighted the positive impact of these AI diagnostic tools on public health, sharing success stories from pilot implementations that illustrate how early and accurate diagnosis has led to better treatment outcomes and reduced disease burden.

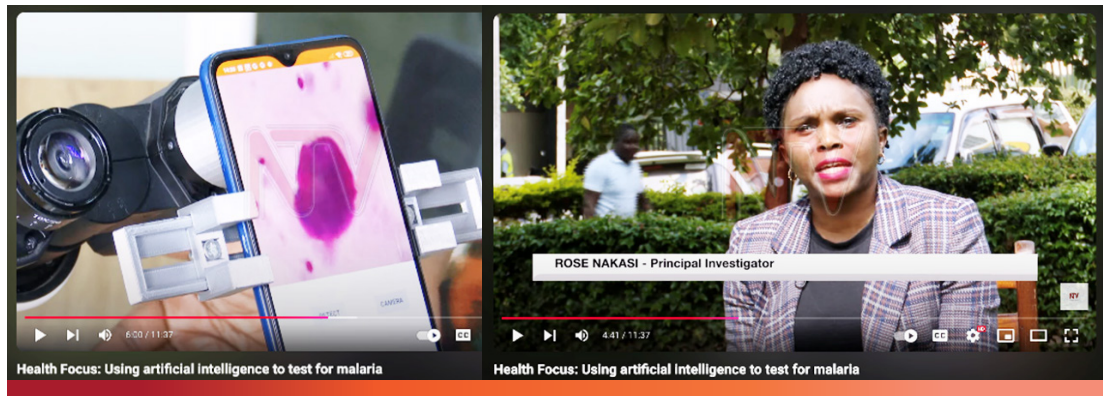
Ocular generated interest and secured significant visibility through prominent media platforms, including:

- **Voice of Africa:** Dr. Rose Nakasi shared about Ocular's innovative mobile microscopy solution during a Voice of Africa segment focused on AI's transformative potential in Africa and its ability

to drive economic growth. She was joined by VOA technology reporter Paul Ndiho and Ambassador Philip Thigo Special envoy on Technology for Kenya



- **Associated Press:** Dr. Rose emphasized that the tool, with appropriate regulations in place, could be particularly beneficial to countries like Uganda in the Global South. By bridging infrastructure and human resource gaps, AI can play a crucial role in addressing healthcare challenges.
- **NTV's "Health Focus":** A feature on Uganda's leading medical segment spotlighted Ocular's AI-driven diagnostic technology. This exposure helped raise public and professional awareness of the project.



Dr. Rose Nakasi's work on automated disease diagnosis featured in the Daily Monitor.

- **Daily Monitor's Healthy Living:** This feature spotlighted Dr. Rose Nakasi's groundbreaking work in utilizing AI for healthcare, specifically showcasing our disease diagnosis tool for common Ugandan diseases like malaria. The segment on Daily Monitor, a leading media outlet, reached a national audience.



Meet Dr Rose Nakasi,

the lead behind an innovative way to screen for malaria

Doing a lot of research, leading an interdisciplinary team and fundraising were some of the challenges Dr. Nakasi faced as she worked on the Mak Ocular project. The project has now received \$1.5 million from Google to advance its work.

By Aaron Oguttu

Artificial intelligence is one of the aspects of technology that has fascinated the world in the 21st

century and keeps bringing up surprises. Meet Dr. Rose Nakasi, a computer scientist with a specialty in Artificial Intelligence (AI) in the Makerere AI Lab. The doctor won a grant worth \$1.5 million (about Shs5.6 billion) from Google to improve the Mak Ocular project she put out. Dr. Nakasi is a lecturer in the Department of Information Technology at Makerere University. She previously lectured at Busitema University. She is currently a lead driver under the global initiative on Artificial Intelligence. She is currently spearheading

[Read more](#)

Ocular featured in Makerere University's The Legacy newsletter

- **Makerere University's "The Legacy" Newsletter:** This feature reached an influential network of alumni, showcasing Ocular's potential to revolutionize healthcare diagnostics, especially in underserved regions.

New Vision: New Vision highlighted our project, showcasing AI's potential not just in diagnosis but across the broader medical field.



The New vision, NBSTV and the Observer on Oculars Google grant

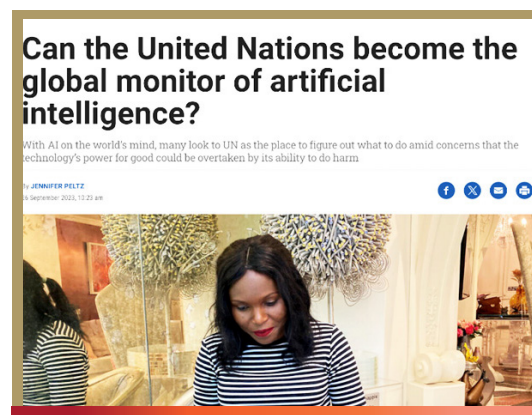


Words That Count: An online publication focusing on STEM-related stories from African women, featured Dr. Rose's journey as an AI in Health researcher. The article highlighted our automated disease diagnosis tool and its potential to develop spatial modeling in a resource-constrained country like Uganda.

Associated Press: Dr. Rose emphasized that the tool, with appropriate regulations in place, could be particularly beneficial to countries like Uganda in the Global South. By bridging infrastructure and human resource gaps, AI can play a crucial role in addressing healthcare challenges



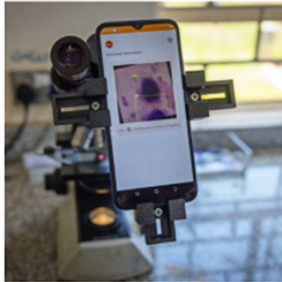
X post by Words that count highlighting Dr. Rose's journey as an AI in Health researcher



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A mobile phone is inserted into a 3D adapter that is attached to the microscope. The phone's camera is used to take pictures of the blood sample, then analyzed and provide a test result in a

They are currently working on AI software that works with any microscope and smartphone, regardless of the model. A pilot project for use across the country is to be developed within three years. At the moment, the focus is still on malaria, but in the future the technology should also be able to be used to diagnose tuberculosis and cervical cancer, explains Nakasi. "This project can improve the performance of the health care system and we hope to get the authorities on board. It helps to combat the shortage of medical professionals if we have technology that can also be used by less qualified staff,"

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Testing our diagnosis tool at Komamboga Health Center

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